

Auction Theory for Business

Strategic Market Design and Applications

Strategic Insights

November 15, 2025

Today's Agenda

Why Should You Care About Auctions?

Auctions are everywhere in business

- Google Ads (\$200+ billion annually, billions of auctions daily)
- Government contracts and procurement
- Spectrum licenses (\$100+ billion raised)
- Real estate and property sales
- Online marketplaces (eBay, stock exchanges)
- Art, wine, and collectibles

Key Business Questions

- Which auction format should I use?
- How do I maximize revenue?
- How should I bid when participating?
- How do I prevent manipulation?

Good market design isn't just theory—it's a competitive advantage.

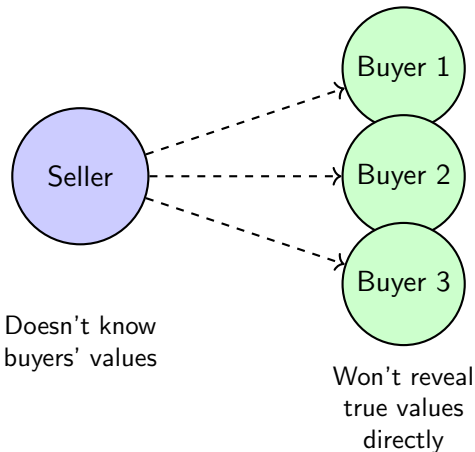
The Stakes Are High

Global Impact Trillions of dollars transacted via auctions annually
Examples of Design Impact

- **UK spectrum auction (2000)** \$35 billion raised with careful design
- **Switzerland spectrum (2000)** Only \$20 million (poor competition)
- **New Zealand disaster** Student bid \$1 for spectrum, won due to no reserve price
- **California energy crisis (2000-01)** \$40+ billion in excess costs from poor design

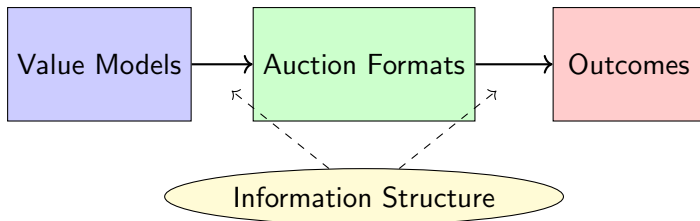
Key Lesson Small design choices = massive financial impact

The Fundamental Challenge



Solution Design an auction that reveals true values through competition

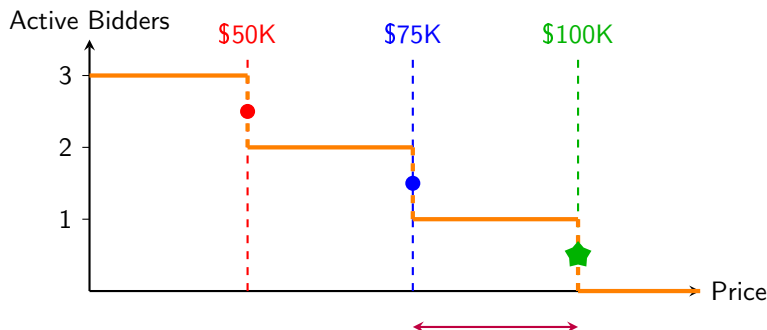
Auction Theory Framework



Key Questions in Auction Theory:

- How does information structure affect bidding?
- Which auction format maximizes revenue?
- When does the winner's curse arise?
- How can sellers use information strategically?

The demand function



Insight 1 If mechanism $\rightarrow p > v_2 \rightarrow b_2 \neq v_2$

Insight 2 Red arrow is efficient price area

Strategic Form Games

Definition: A game $\Gamma = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ consists of:

- **Players:** $N = \{1, 2, \dots, n\}$
- **Strategy spaces:** S_i for each player i
- **Payoff functions:** $u_i : S_1 \times \dots \times S_n \rightarrow \mathbb{R}$

Solution Concepts:

- ① **Dominant Strategy:** $s_i^* \in S_i$ such that

$$u_i(s_i^*, s_{-i}) \geq u_i(s_i, s_{-i}) \quad \forall s_i \in S_i, \forall s_{-i} \in S_{-i}$$

- ② **Nash Equilibrium:** Profile $(s_1^*, \dots, s_n^*)$ where

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in S_i, \forall i \in N$$

Incomplete Information Games

Harsanyi Transformation:

- Nature determines types $\theta = (\theta_1, \dots, \theta_n)$
- Player i observes only θ_i
- Common prior: $\theta \sim F$ where F is common knowledge
- Strategy: $s_i : \Theta_i \rightarrow S_i$

Bayesian Nash Equilibrium:

Strategy profile $(s_1^*, \dots, s_n^*)$ where each $s_i^* : \Theta_i \rightarrow S_i$ satisfies:

$$s_i^*(\theta_i) \in \arg \max_{s_i} \mathbb{E}_{\theta_{-i}|\theta_i}[u_i(s_i, s_{-i}^*(\theta_{-i}), \theta_i, \theta_{-i})]$$

Key Insight: Each type of each player best responds to beliefs about others' strategies

Order Statistics: Definition and Notation

Setup: Given n i.i.d. random variables $X_1, X_2, \dots, X_n \sim F$

Order Statistics: Rearrange in ascending order

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

Alternative Notation (Auctions): Often use descending order

$$v_{(1)} \geq v_{(2)} \geq \dots \geq v_{(n)}$$

where $v_{(1)} = \max\{v_1, \dots, v_n\}$ and $v_{(n)} = \min\{v_1, \dots, v_n\}$

Why Important for Auctions?

- Winner has highest valuation: $v_{(1)}$
- Second-price auctions: Winner pays $v_{(2)}$
- Revenue comparisons require $\mathbb{E}[v_{(1)}]$ and $\mathbb{E}[v_{(2)}]$

Distribution of Order Statistics

CDF of k -th Order Statistic:

The probability that the k -th smallest value is at most x :

$$G_{(k)}(x) = \Pr(X_{(k)} \leq x) = \sum_{j=k}^n \binom{n}{j} F(x)^j [1 - F(x)]^{n-j}$$

Intuition: At least k of the n values must be $\leq x$

PDF of k -th Order Statistic:

$$g_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1} [1 - F(x)]^{n-k} f(x)$$

Special Cases:

- Maximum: $G_{(n)}(x) = F(x)^n$, $g_{(n)}(x) = nF(x)^{n-1}f(x)$
- Minimum: $G_{(1)}(x) = 1 - (1 - F(x))^n$, $g_{(1)}(x) = n(1 - F(x))^{n-1}f(x)$

First and Second Order Statistics

First Order Statistic (Maximum):

- Distribution: $G_{(1)}(v) = F(v)^n$
- Density: $g_{(1)}(v) = nF(v)^{n-1}f(v)$
- Expected value: $\mathbb{E}[v_{(1)}] = \int_{\underline{v}}^{\bar{v}} v \cdot nF(v)^{n-1}f(v)dv$

Second Order Statistic:

- Distribution: $G_{(2)}(v) = nF(v)^{n-1}[1 - F(v)] + F(v)^n$
- Density: $g_{(2)}(v) = n(n-1)F(v)^{n-2}[1 - F(v)]f(v)$
- Expected value: $\mathbb{E}[v_{(2)}] = \int_{\underline{v}}^{\bar{v}} v \cdot g_{(2)}(v)dv$

Useful Identity:

$$\mathbb{E}[v_{(2)}] = n\mathbb{E}[v_{(1,n-1)}] - (n-1)\mathbb{E}[v_{(1,n)}]$$

where $v_{(1,m)}$ denotes the maximum of m draws

Uniform Distribution: Key Results

For $U[0, 1]$ distribution:

General Formula:

$$\mathbb{E}[X_{(k)}] = \frac{k}{n+1}$$

Intuition: Order statistics divide $[0, 1]$ into $n+1$ equal segments in expectation

Examples:

- $n = 2$: $\mathbb{E}[v_{(1)}] = \frac{2}{3}$, $\mathbb{E}[v_{(2)}] = \frac{1}{3}$
- $n = 3$: $\mathbb{E}[v_{(1)}] = \frac{3}{4}$, $\mathbb{E}[v_{(2)}] = \frac{2}{4} = \frac{1}{2}$
- $n = 10$: $\mathbb{E}[v_{(1)}] = \frac{10}{11}$, $\mathbb{E}[v_{(2)}] = \frac{9}{11}$

Variance:

$$\text{Var}(X_{(k)}) = \frac{k(n-k+1)}{(n+1)^2(n+2)}$$

Order Statistics: Computational Examples

Example 1: Two Dice Rolls

- Roll two dice, let $V_{(1)} \geq V_{(2)}$ be the ordered outcomes
- Distribution of maximum:

v	1	2	3	4	5	6
$\Pr(V_{(1)} = v)$	1/36	3/36	5/36	7/36	9/36	11/36

- $\mathbb{E}[V_{(1)}] = 161/36 \approx 4.47$
- $\mathbb{E}[V_{(2)}] = 91/36 \approx 2.53$

Example 2: Exponential Distribution

- If $X_i \sim \text{Exp}(\lambda)$ i.i.d., then:
- $X_{(1)} \sim \text{Exp}(n\lambda)$ (minimum is also exponential)
- $\mathbb{E}[X_{(1)}] = \frac{1}{n\lambda}$

Types of Information in Auctions

1. Independent Private Values (IPV):

- Each bidder knows own valuation exactly: v_i
- Values are private information, drawn i.i.d. from F
- Learning others' values doesn't change own value
- Example: Personal use items (used car, collectible)

2. Common Values (CV):

- Single unknown value V same for all bidders
- Each bidder receives private signal s_i about V
- Winner's curse: Winning means you had highest estimate
- Example: Oil drilling rights, company takeovers

3. Interdependent/Affiliated Values:

- General case: $v_i = v_i(s_1, \dots, s_n)$
- Signals may be correlated (affiliated)
- Encompasses both IPV and CV as special cases

Understanding Value Types

Private Values value is independent of others

- Personal consumption items
- Office space for your own use
- Company car for internal use

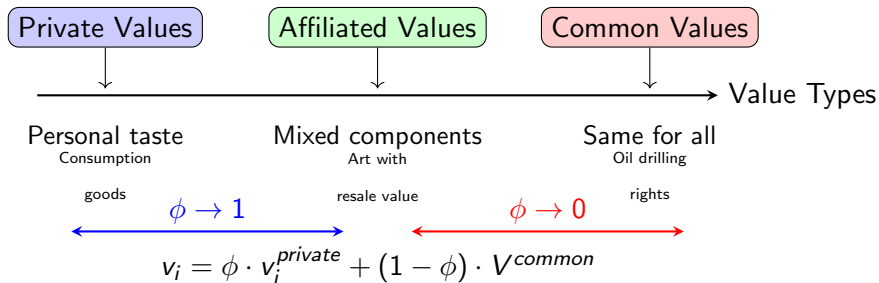
Common Values value is the same for everyone

- Oil drilling rights (unknown amount of oil)
- Company acquisition (unknown future profits)
- Government bonds

Mixed Values

- Real estate (personal use + investment value)
- Art (enjoyment + resale value)

Visualization: Value Types Spectrum



Common Knowledge and Beliefs

Common Knowledge:

Event E is common knowledge if:

- ① Everyone knows E
- ② Everyone knows that everyone knows E
- ③ Everyone knows that everyone knows that everyone knows E
- ④ ... (ad infinitum)

In Auctions, Common Knowledge Includes:

- Number of bidders n
- Auction format and rules
- Distribution F (but not realizations)
- Rationality of all bidders
- Reserve price (if public)

Private Information:

- Own valuation or signal
- Own bidding strategy (in sealed-bid)
- Others' valuations or signals

Standard Auction Formats

Open Auctions:

- **English (Ascending):** Price rises, bidders drop out
- **Dutch (Descending):** Price falls, first to accept wins

Sealed-Bid Auctions:

- **First-Price:** Highest bid wins, pays own bid
- **Second-Price (Vickrey):** Highest bid wins, pays second-highest bid

Strategic Equivalences (IPV):

- Dutch $\equiv$ First-Price Sealed-Bid
- English $\approx$ Second-Price Sealed-Bid

Dominant Strategies:

- Second-Price: Bid true value (truthful)
- English: Stay in until price exceeds value
- First-Price/Dutch: No dominant strategy (bid shading)

Bayesian Nash Equilibrium in Auctions

Definition: Strategy profile $\beta^* = (\beta_1^*, \dots, \beta_n^*)$ is a BNE if:

$$\beta_i^*(v_i) \in \arg \max_b \mathbb{E}_{v_{-i}}[u_i(b, \beta_{-i}^*(v_{-i}), v_i)]$$

Symmetric Equilibrium:

- All bidders use same strategy: $\beta_i = \beta$ for all i
- Strategy is strictly increasing in valuation
- Boundary condition: $\beta(\underline{v}) = \underline{v}$ (or 0)

First-Price Auction Equilibrium (IPV):

$$\beta(v) = v - \frac{\int_{\underline{v}}^v F(t)^{n-1} dt}{F(v)^{n-1}}$$

Uniform $[0, 1]$ Case:

$$\beta(v) = \frac{n-1}{n} v$$

Revenue Equivalence Theorem

Theorem (Vickrey 1961, Riley-Samuelson 1981):

Under IPV with risk-neutral bidders and symmetric equilibrium:

- All standard auctions yield same expected revenue
- Expected revenue = $\mathbb{E}[v_{(2)}]$ (second-order statistic)

Conditions for Revenue Equivalence:

- 1 Independent private values
- 2 Risk neutrality
- 3 Symmetric bidders
- 4 Same allocation rule (efficient)
- 5 Same payment by lowest type

Breaking Revenue Equivalence:

- Risk aversion $\Rightarrow$ First-price dominates
- Affiliated values $\Rightarrow$ English dominates
- Asymmetric bidders $\Rightarrow$ No general ranking

Differential Equations in Auction Theory

First-Order Approach:

For symmetric equilibrium $\beta(v)$ in first-price auction:

- 1 Bidder's problem: $\max_b (v - b) \cdot \Pr(\text{win with } b)$
- 2 FOC at equilibrium: Leads to differential equation
- 3 General form: $\beta'(v) = h(v)(v - \beta(v))$
- 4 Boundary condition: $\beta(\underline{v}) = \underline{v}$ or reserve price

Solution Technique:

- Rewrite as: $d[\beta(v)H(v)] = v \cdot dH(v)$
- Integrate with boundary condition
- Verify second-order conditions

Example - Uniform Distribution:

$$\beta'(v) = \frac{(n-1)(v - \beta(v))}{v} \Rightarrow \beta(v) = \frac{n-1}{n}v$$

Integration by Parts in Revenue Calculations

Expected Revenue Formula:

$$\mathbb{E}[\text{Revenue}] = \int_{\underline{v}}^{\bar{v}} \beta(v) \cdot g_{(1)}(v) dv$$

Key Technique: Integration by parts

$$\begin{aligned} \int_{\underline{v}}^{\bar{v}} v \cdot dF^n(v) &= [v \cdot F^n(v)]_{\underline{v}}^{\bar{v}} - \int_{\underline{v}}^{\bar{v}} F^n(v) dv \\ &= \bar{v} - \int_{\underline{v}}^{\bar{v}} F^n(v) dv \end{aligned}$$

Useful Identity for Second-Order Statistic:

$$\mathbb{E}[v_{(2)}] = \int_{\underline{v}}^{\bar{v}} v \cdot n(n-1)F^{n-2}(1-F)f dv$$

Revenue Equivalence Check: All standard auctions yield

$$\mathbb{E}[\text{Revenue}] = \mathbb{E}[v_{(2)}]$$

Reserve Prices and Optimal Auctions

Optimal Reserve Price:

Maximizes expected revenue subject to participation constraints

Myerson (1981) Result:

$$r^* = v_s + \frac{1 - F(r^*)}{f(r^*)}$$

where v_s is seller's valuation

Virtual Valuation:

$$\psi(v) = v - \frac{1 - F(v)}{f(v)}$$

- Optimal auction allocates to bidder with highest positive virtual valuation
- Reserve price where $\psi(r^*) = 0$

Example - Uniform $[0, 1]$:

- Virtual valuation: $\psi(v) = 2v - 1$
- Optimal reserve: $r^* = 1/2$
- Independent of number of bidders!

All-Pay Auctions and Contests

All-Pay Auction:

- Everyone pays their bid, only highest wins prize
- Models: R&D races, political lobbying, litigation

Equilibrium Bidding (Symmetric IPV):

$$\beta_{AP}(v) = \int_{\underline{v}}^v t \cdot dF^{n-1}(t) = \mathbb{E}[v_{(1,n-1)} | v_{(1,n-1)} < v]$$

Uniform $[0, 1]$ Case:

$$\beta_{AP}(v) = \frac{n-1}{n} v^n$$

Revenue:

- Total revenue = $n \cdot \mathbb{E}[\beta_{AP}(v)]$ (all pay)
- Still equals $\mathbb{E}[v_{(2)}]$ under revenue equivalence!

Common Values and Winner's Curse

Winner's Curse:

- In CV auctions, winning is “bad news” about value
- Winner had most optimistic signal
- Naive bidding leads to losses

Equilibrium Adjustment:

Rational bidders account for adverse selection:

$$\beta(s_i) = \mathbb{E}[V|s_i, \text{Win}] < \mathbb{E}[V|s_i]$$

Mineral Rights Model:

- True value: $V = \frac{1}{n} \sum_{i=1}^n s_i$
- Signals: $s_i = V + \epsilon_i$, ϵ_i i.i.d.
- More bidders $\Rightarrow$ Worse winner's curse
- English auction reveals information, reduces curse

Linkage Principle: More information revelation increases expected revenue

What Are We Studying? The Auction Game

The Basic Auction Story:

- Multiple people want the same thing (a house, a painting, ad space)
- Only one can have it (or limited quantities available)
- How do we decide who gets it and at what price?

Why Game Theory?

- Your best strategy depends on what others do
- Others' strategies depend on what you do
- Everyone is trying to outsmart everyone else
- Information is often incomplete or asymmetric

The Central Questions:

- How should rational bidders behave?
- What auction format maximizes seller revenue?
- How does information structure affect outcomes?
- When do auctions fail to work well?

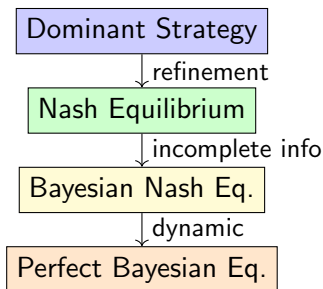
Think of auctions as strategic battlegrounds where information, timing, and psychology all matter!

Game-Theoretic Foundations

Strategic Form Game: $G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$

- Players: $N = \{1, 2, \dots, n\}$
- Strategy spaces: S_i for player i
- Payoff functions: $u_i : S_1 \times \dots \times S_n \rightarrow \mathbb{R}$

Solution Concepts Hierarchy:



Order Statistics: Essential Theory

Definition: For i.i.d. $X_1, \dots, X_n \sim F$, order statistics are:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

Distribution of k -th Order Statistic:

$$F_{(k)}(x) = \sum_{j=k}^n \binom{n}{j} F(x)^j [1 - F(x)]^{n-j}$$

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1} [1 - F(x)]^{n-k} f(x)$$

Key Result for Auctions

For uniform $[0, 1]$ distribution:

$$\mathbb{E}[X_{(k)}] = \frac{k}{n+1}, \quad \text{Var}(X_{(k)}) = \frac{k(n-k+1)}{(n+1)^2(n+2)}$$

Game Theory Takeaways

What We Learned:

- **Strategic Thinking:** In auctions, you must think about what others are thinking
- **Solution Concepts:** Different levels of rationality lead to different predictions
- **Order Statistics:** The mathematics of "who has the highest value" is crucial

Key Insights:

- Dominant strategies (like truthful bidding in second-price) are rare but powerful
- Most auction analysis requires equilibrium thinking
- The second-highest value matters as much as the highest!

Why This Matters:

- Foundation for understanding bidding strategies
- Explains why different auction formats exist
- Shows how information and beliefs drive behavior

Why Knowledge Matters in Auctions

The Information Puzzle:

- In auctions, what you know matters
- But what you know that others know also matters
- And what you know that others know that you know...
- This infinite regress is crucial for strategic reasoning!

Common Knowledge in Practice:

- Auction rules must be common knowledge
- Reserve prices are often announced publicly (why?)
- Some information is deliberately kept private
- Strategic information revelation can increase revenue

Knowledge isn't just about facts—it's about knowing what others know!

Knowledge Operators: Philosophical Foundations

Epistemic Logic Framework:

- Knowledge operator K_i : “agent i knows that...”
- $K_i\varphi$ means “agent i knows proposition φ ”

S5 Axiom System (Kripke, 1963):

- 1 **Knowledge:** $K_i\varphi \rightarrow \varphi$ (truth axiom)
- 2 **Positive Introspection:** $K_i\varphi \rightarrow K_iK_i\varphi$
- 3 **Negative Introspection:** $\neg K_i\varphi \rightarrow K_i\neg K_i\varphi$
- 4 **Distribution:** $K_i(\varphi \rightarrow \psi) \rightarrow (K_i\varphi \rightarrow K_i\psi)$
- 5 **Necessitation:** If $\vdash \varphi$, then $\vdash K_i\varphi$

Philosophical Implications:

- Agents have perfect knowledge of their own knowledge state
- No false beliefs (knowledge implies truth)
- Logical omniscience problem: agents know all tautologies

Kripke Structures and Possible Worlds

Formal Model:

- State space Ω (possible worlds)
- Accessibility relation $\sim_i \subseteq \Omega \times \Omega$ for each agent i
- Valuation function $V : \Omega \times \Phi \rightarrow \{0, 1\}$

Knowledge Definition:

$$(\Omega, \omega) \models K_i \varphi \iff \forall \omega' \in \Omega : \omega \sim_i \omega' \implies (\Omega, \omega') \models \varphi$$

Properties of $\sim_i$ in S5:

- **Reflexive:** $\omega \sim_i \omega$ (truth axiom)
- **Symmetric:** $\omega \sim_i \omega' \implies \omega' \sim_i \omega$
- **Transitive:** $\omega \sim_i \omega' \wedge \omega' \sim_i \omega'' \implies \omega \sim_i \omega''$

Therefore, $\sim_i$ forms an equivalence relation (information partition)

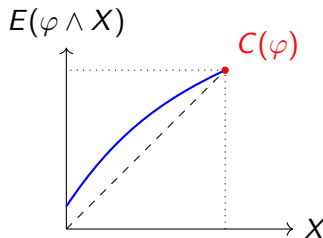
Iterative Common Knowledge and Aumann(1976)

$$E^1(\varphi) = \bigwedge_{i \in N} K_i \varphi \quad (\text{everyone knows})$$

$$E^k(\varphi) = E(E^{k-1}(\varphi)) \quad (\text{k-th level mutual knowledge})$$

$$C(\varphi) = \bigwedge_{k=1}^{\infty} E^k(\varphi) \quad (\text{common knowledge})$$

$$C(\varphi) \leftrightarrow E(\varphi \wedge C(\varphi)) \quad (\text{Fixed Point})$$



The Coordinated Attack Problem

Byzantine Generals Paradox (Philosophical Insight):

- Two generals must coordinate attack
- Messages may be lost
- Common knowledge impossible with unreliable communication

Formal Result:

Theorem

In a system with possible communication failure, common knowledge of any non-trivial fact cannot be achieved in finite time.

Auction Theory Implication:

- Perfect common knowledge is idealization
- Approximate common knowledge: ε -common belief
- Robust mechanism design accounts for higher-order uncertainty

Knowledge and Information: Key Lessons

What We Learned:

- **Knowledge Hierarchy:** "I know" vs "We all know that we all know"
- **Common Knowledge:** The infinite mutual knowledge that enables coordination
- **Practical Limits:** Perfect common knowledge is impossible in real markets

Real-World Applications:

- eBay shows bid history (creates common knowledge)
- Art auctions use elaborate signaling systems
- Financial markets have designated information releases

Remember: In auctions, it's not just what you know—it's what everyone knows that everyone knows!

Understanding Information in Auctions

Private Values Example - Used Car:

- You know how much you need a car for commuting
- Others' valuations don't change your needs
- Your value is certain and private

Common Values Example - Oil Drilling Rights:

- Nobody knows exactly how much oil is there
- Everyone gets different geological surveys (signals)
- The actual amount of oil is the same for everyone
- Learning others' signals would change your estimate

The information structure determines everything: bidding strategies, revenue, efficiency, and even whether the auction works at all!

Type Spaces and Beliefs

Harsanyi Type Space $(T_i, \tau_i, p_i)_{i \in N}$:

- T_i : Type space for player i
- $\tau_i : \Theta \rightarrow T_i$: Type function mapping states to types
- $p_i : T_i \rightarrow \Delta(T_{-i} \times \Theta)$: Belief function

Consistency Requirements:

- 1 **Own-type awareness:** $p_i(t_i)(\{t_{-i}\} \times \{\theta\}) > 0 \implies \tau_i(\theta) = t_i$
- 2 **Common prior (optional):** $\exists \mu \in \Delta(\Theta)$ such that beliefs derive from Bayesian updating

Universal Type Space (Mertens-Zamir, 1985):

- Contains all possible belief hierarchies
- $T_i^* = \prod_{k=1}^{\infty} \Delta^k$ where Δ^k is k -th order beliefs
- Every finite type space embeds into universal space

Information Structures in Auctions

Private Values Model (PV):

- Value: $v_i = \theta_i$ (private information = value)
- Type space: $T_i = [\underline{v}, \bar{v}]$
- Common prior: $v_i \sim F$ i.i.d.
- Utility: $u_i(x, p, v_i) = v_i \cdot x_i - p_i$

Common Values Model (CV):

- True value: $V(\theta)$ where $\theta = (s_1, \dots, s_n)$
- Signal: $s_i \in S_i$ privately observed
- Value function: Often $V = \frac{1}{n} \sum_{i=1}^n s_i$ or $V = \max_i s_i$
- Utility: $u_i(x, p, \theta) = V(\theta) \cdot x_i - p_i$

Interdependent Values (IV):

- General case: $v_i(s_1, \dots, s_n)$
- Nests PV ($v_i(s) = s_i$) and CV ($v_i(s) = V(s) \forall i$)
- Single Crossing: $\frac{\partial^2 v_i}{\partial s_i \partial s_j} \geq 0$ (affiliation)

Six auction Formats

Let v = your valuation, b = your bid, b_1 = highest bid, b_2 = second-highest bid

① **English Auction (Ascending Price)**

② **Dutch Auction (Descending Price)**

③ **Sealed-Bid First-Price**

$$\pi = \mathbb{P}(b = b_1) \cdot (v - b)$$

④ **Sealed-Bid Second-Price (Vickrey)**

$$\pi = \mathbb{P}(b = b_1) \cdot (v - b_2)$$

⑤ **All-Pay First Price**

$$\pi = \mathbb{P}(b = b_1) \cdot v - b$$

⑥ **All-Pay Second Price (War of Attrition)**

$$\pi = \mathbb{P}(b = b_1) \cdot (v - b_2) - \mathbb{P}(b \neq b_1) \cdot b$$

What is an English Auction?

Definition

An **English auction** is a market mechanism where bidders **sequentially** indicate their interest in purchasing a product at a current price. Only the winner pays.

Standard Ascending Format

- Prices **increase** incrementally
- Bidders signal their interest
- Ends when only one bidder signals

Reverse English Format

- Prices **decrease** incrementally
- Bidders signal interest
- Ends when only one bidder signals

Note

English = Ascending; **Reverse** Ascending = **Reverse** English The terminology refers to the auction format, not the price direction.

English Auction: Equilibrium Analysis

Dominant Strategy:

- Stay in auction while price $p < v_i$
- Drop out when price $p \geq v_i$
- Strategy independent of beliefs about others

Equilibrium Outcome:

- Bidder 2 drops out at $p = v_{(2)}$
- Bidder 1 (highest value) wins
- Pays price $p = v_{(2)}$ (second-highest valuation)
- Winner's surplus: $v_{(1)} - v_{(2)} > 0$

Properties:

- Pareto efficient (highest value wins)
- Monopolist cannot extract full surplus
- Simple strategy (no need to know F or n)
- Transparent process

Derivation: English Auction

Claim: Staying active while $p < v_i$ is weakly dominant

Proof: Consider bidder i with value v_i , current price p

Case 1: $p < v_i$

- Stay active: Win if others drop first, payoff $= v_i - p_{final} \geq v_i - p > 0$
- Drop out: Lose, payoff $= 0$
- Staying dominates dropping

Case 2: $p \geq v_i$

- Stay active: If win, pay $p \geq v_i$, payoff ≤ 0
- Drop out: Lose, payoff $= 0$
- Dropping (weakly) dominates staying

Equilibrium Bidding Function:

$$b(v_i) = v_i \text{ (stay active until price reaches } v_i \text{)}$$

Winner pays: $p^* = v_{(2)}$ (second-highest valuation)

What is a Dutch Auction?

Definition

A **Dutch auction** is a market mechanism where an auctioneer **sequentially** changes the price until the first person signals their interest. The first bidder to accept, wins.

Standard Descending Format

- Prices **decrease** incrementally
- Bidders watch and wait
- First to accept the price wins

Reverse Dutch

- Prices **increase** incrementally
- Bidders watch and wait
- First to accept the price wins

Note

Reverse Descending = **Reverse** Dutch; Descending = Dutch
The terminology refers to the auction format, not the price direction.

Dutch Auction: Strategic Equivalence

Claim: Dutch and First-Price auctions are strategically equivalent

Dutch Auction Decision:

- Choose stopping price b before auction starts
- Accept when clock reaches b
- If someone else accepts first, lose

First-Price Decision:

- Choose bid b to submit in sealed envelope
- If b is highest, win and pay b
- Otherwise lose

Why Equivalent?

- Same information at decision time (only own value)
- Same action space (choose a price)
- Same payoff structure
- Same equilibrium strategies

Derivation: Dutch Auction

Setup:

- Clock starts at high price, descends continuously
- First bidder to accept wins at current price

Strategic Equivalence to First-Price:

Dutch Decision: Choose stopping price b ex-ante

- Accept when clock reaches b
- Win if no one accepts earlier (i.e., $b > b_j$ for all $j \neq i$)
- Pay b if win
- Payoff: $(v_i - b) \cdot \mathbb{P}(b > \max_{j \neq i} b_j)$

First-Price Decision: Submit sealed bid b

- Win if $b > b_j$ for all $j \neq i$
- Pay b if win
- Payoff: $(v_i - b) \cdot \mathbb{P}(b > \max_{j \neq i} b_j)$

Conclusion: Identical payoff structures $\Rightarrow$ identical equilibrium strategies

$$b_{Dutch}(v_i) = b_{First}(v_i) = \mathbb{E}[v_{(2)} \mid v_{(1)} = v_i]$$

What is a First Price Auction?

Definition

A **first price auction** is a **simultaneous** market mechanism where the highest bidder wins and pays their own bid.

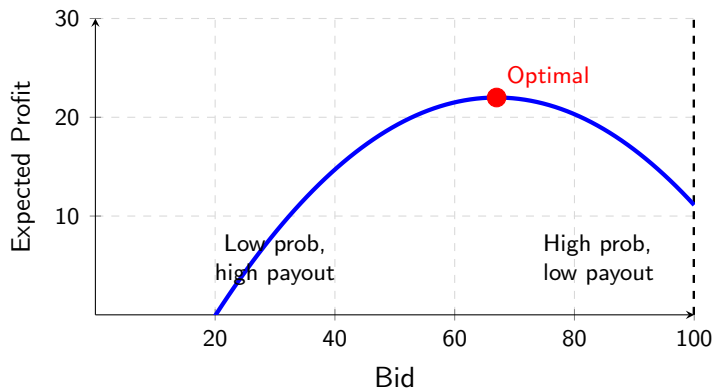
Sealed-Bid First Price Format

- Bidders submit sealed bids
- Highest bid wins the item
- Winner pays their own bid

Note

In a first price auction, bidders face a trade-off bidding high increases the chance of winning but decreases profit if they win. This is known as the bid-shading problem.

Sealed-Bid First price The Strategic Challenge



The Trade-off Higher bid increases win chance but reduces profit

First-Price: Equilibrium Derivation

Maximization Problem: If bidder i with value v_i bids as if value w_i :

$$\mathbb{E}[U_i] = (v_i - B(w_i)) \cdot F^{n-1}(w_i)$$

First-Order Condition:

$$\frac{\partial}{\partial w_i} : (v_i - B(w_i))(n-1)f(w_i)F^{n-2}(w_i) - B'(w_i)F^{n-1}(w_i) = 0$$

At symmetric equilibrium $w_i = v_i$:

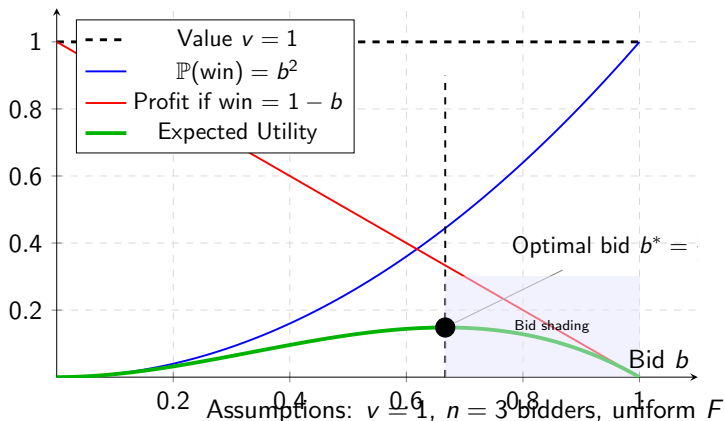
$$B'(v_i) = (n-1)(v_i - B(v_i)) \frac{f(v_i)}{F(v_i)}$$

General Solution: (with $B(\underline{v}) = \underline{v}$)

$$B(v_i) = \frac{\int_{\underline{v}}^{v_i} t dF^{n-1}(t)}{F^{n-1}(v_i)} = v_i - \frac{\int_{\underline{v}}^{v_i} F^{n-1}(t) dt}{F^{n-1}(v_i)}$$

Note: $B(v_i) < v_i$ (always shade bid)

Visualization: First-Price Bid Shading Trade-off



First-Price: Equilibrium Derivation

Maximization Problem: If bidder i with value v_i bids as if value w_i :

$$\mathbb{E}[U_i] = (v_i - B(w_i)) \cdot F^{n-1}(w_i)$$

First-Order Condition:

$$\frac{\partial}{\partial w_i} : (v_i - B(w_i))(n-1)f(w_i)F^{n-2}(w_i) - B'(w_i)F^{n-1}(w_i) = 0$$

At symmetric equilibrium $w_i = v_i$:

$$B'(v_i) = (n-1)(v_i - B(v_i)) \frac{f(v_i)}{F(v_i)}$$

General Solution: (with $B(\underline{v}) = \underline{v}$)

$$B(v_i) = \frac{\int_{\underline{v}}^{v_i} t dF^{n-1}(t)}{F^{n-1}(v_i)} = v_i - \frac{\int_{\underline{v}}^{v_i} F^{n-1}(t) dt}{F^{n-1}(v_i)}$$

Note: $B(v_i) < v_i$ (always shade bid)

First-Price: Uniform Distribution Example

Setup: $v \sim U[0, 1]$, so $F(v) = v$, $f(v) = 1$

Differential equation:

$$B'(v_i) = \frac{(n-1)(v_i - B(v_i))}{v_i}$$

Guess linear solution: $B(v_i) = kv_i$

Substituting: $k = \frac{(n-1)(v_i - kv_i)}{v_i} = (n-1)(1-k)$

Solving: $k = \frac{n-1}{n}$

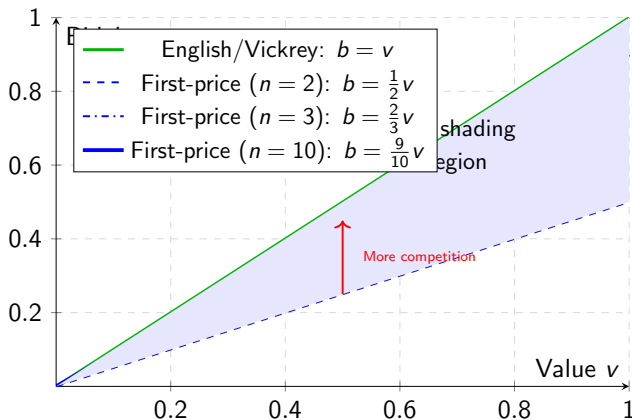
Equilibrium bid function:

$$B(v_i) = \frac{n-1}{n} v_i$$

Interpretation:

- 2 bidders: bid 50% of value
- 10 bidders: bid 90% of value
- As $n \rightarrow \infty$: $B(v) \rightarrow v$ (competition eliminates shading)

Visualization: Bid Functions Across Auction Types



Derivation: First-Price Auction (Detailed)

$v_i \sim F(v) - [\underline{v}, \bar{v}]$, symmetric equilibrium $B(v)$ bids as if value w :

$$U(w; v_i) = (v_i - B(w)) \cdot F^{n-1}(w)$$

$$\frac{\partial U}{\partial w} = -B'(w)F^{n-1}(w) + (v_i - B(w))(n-1)f(w)F^{n-2}(w) = 0$$

Symmetric Equilibrium ($w = v_i$) :

$$B'(v_i)F^{n-1}(v_i) = (v_i - B(v_i))(n-1)f(v_i)F^{n-2}(v_i)$$

$$\rightarrow B'(v_i) = (n-1)(v_i - B(v_i)) \frac{f(v_i)}{F(v_i)}$$

Solution (first-order linear ODE with $B(\underline{v}) = \underline{v}$) :

$$B(v_i) = \frac{\int_{\underline{v}}^{v_i} t dF^{n-1}(t)}{F^{n-1}(v_i)} = v_i - \frac{\int_{\underline{v}}^{v_i} F^{n-1}(t) dt}{F^{n-1}(v_i)}$$

Derivation: First-Price Uniform Example

Setup: $v_i \sim U[0, 1]$, so $F(v) = v$, $f(v) = 1$

$$\text{ODE } B'(v_i) = (n-1)(v_i - B(v_i)) \frac{1}{v_i}$$

Guess Linear Solution: $B(v_i) = \alpha v_i$ for some $\alpha \in [0, 1]$

$$\alpha = (n-1)(v_i - \alpha v_i) \frac{1}{v_i} = (n-1)(1 - \alpha)$$

$$\rightarrow \alpha = \frac{n-1}{n}$$

$$\rightarrow \boxed{B(v_i) = \frac{n-1}{n} v_i}$$

Verification: Substitute back into ODE to confirm it satisfies the differential equation and boundary condition $B(0) = 0$. ✓

What is a Second Price Auction?

Definition

A **second price auction** is a **simultaneous** market mechanism where the highest bidder wins but pays the price bid by the second-highest bidder.

Sealed-Bid Second Price Format

- Bidders submit sealed bids
- Highest bid wins the item
- Winner pays the second-highest bid

Note

In a second price auction, bidding your true valuation is optimal. There is no incentive to bid strategically lower than your actual willingness to pay.

Second Price Dominant Strategy Theorem

Theorem (Dominant Strategy)

In a second-price sealed-bid auction, bidding true valuation $b_i = v_i$ is a weakly dominant strategy for all bidders.

Proof Let $\tilde{b} = \max_{j \neq i} b_j$ (highest other bid)

Case 1 Overbidding ($b > v$)

- If $\tilde{b} > v$, then you don't want to win
- If $\tilde{b} < v$, you can also win by bidding $b = v$, payoff is same $v - \tilde{b} > 0$

Bidding v dominates avoids losses, keeps all gains

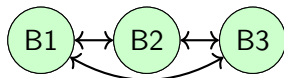
Case 2 Underbidding ($b < v$)

- If $v > \tilde{b} > b$ Lose, but could win profitably by bidding higher
- If $v > b > \tilde{b}$ Win with same profit as bidding v

Bidding v dominates captures all profitable opportunities $\square$

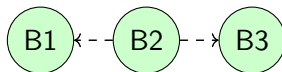
Information Flow in Different Formats

English Auction



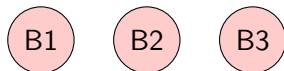
Full information flow

Second-Price



Winner learns 2nd bid

First-Price



No information flow

Information Revelation Impact:

- More information → Less winner's curse
- Less winner's curse → More aggressive bidding
- More aggressive bidding → Higher revenue

What is an All-Pay Auction?

Definition

An **all-pay auction** is a **simultaneous** market mechanism where the highest bidder wins the item, but **all bidders pay their bids** regardless of whether they win or lose.

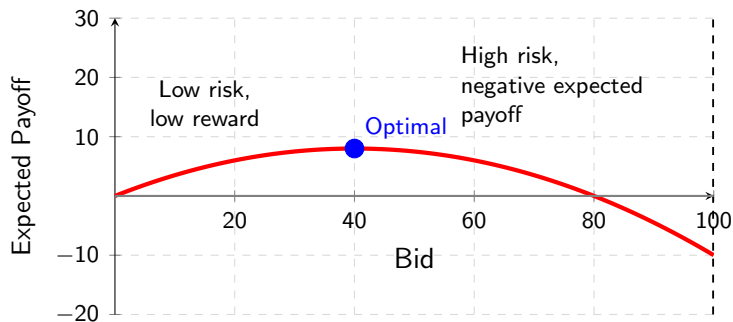
Sealed-Bid All-Pay Format

- Bidders submit sealed bids
- Highest bid wins the item
- **All bidders pay their bid** (winner and losers)
- Winner receives the item, losers receive nothing

Note

All-pay auctions model situations like lobbying, R&D races, and contests where effort is expended regardless of success. Bidders bid much more conservatively than in standard auctions since losing is costly.

All-Pay Auction: The Strategic Challenge



Key Insight: Since you pay even if you lose, optimal bids are much lower than in standard auctions. Risk of loss constrains bidding behavior.

All-Pay Auctions in Practice: Lobbying and Contests

Real-World Applications

- **Political lobbying:** Campaign contributions spent regardless of outcome
- **R&D races:** Research costs incurred by all firms, only winner profits
- **Legal battles:** Legal fees paid by all parties, winner takes all
- **Sports tournaments:** Training costs for all athletes, prizes only for winners

Example: Pharmaceutical R&D

R&D Spending	Outcome
Company A: \$500M	Patent granted
Company B: \$400M	No patent
Company C: \$300M	No patent
Company D: \$200M	No patent

Strategic Implication: Firms must balance investment level against probability of success, knowing sunk costs are irrecoverable.

All-Pay Auctions

Rules:

- Highest bidder wins with certainty
- All bidders pay their bids (non-refundable)
- Applications: Patent races, political lobbying

Symmetric Equilibrium:

$$b(v_i) = \int_{\underline{v}}^{v_i} t dF^{N-1}(t) = v_i F^{N-1}(v_i) - \int_{\underline{v}}^{v_i} F^{N-1}(t) dt$$

Uniform Distribution Example: $v \sim U[0, 1]$

$$b(v_i) = \frac{N-1}{N} v_i^N$$

Compare to first-price auction: $b_{FPA}(v_i) = \frac{N-1}{N} v_i$

Note: $b_{FPA}(v_i) > b_{All-Pay}(v_i)$ for all $v_i < 1$

Revenue Equivalence: $ER_{All-Pay} = E(v_{(2)})$

War of Attrition

Second-Price All-Pay Auction:

- Highest bidder wins
- Winner pays second-highest bid
- All losers pay their own bids
- Application: Animal contests, standards wars

Equilibrium Strategy:

$$b(v_i) = -v_i \ln(1 - F^{N-1}(v_i)) + \int_{\underline{v}}^{v_i} \ln(1 - F^{N-1}(t)) dt$$

Two-Player Uniform Example:

$$b(v_i) = -v_i - \ln(1 - v_i)$$

Key property: $\lim_{v_i \rightarrow 1} b(v_i) = \infty$ (highest type bids infinity!)

Revenue with Affiliation:

$$ER_{War} \geq ER_{2nd-All-Pay} \geq ER_{1st-Price} = ER_{Dutch}$$

Derivation: All-Pay First-Price Auction

Expected Utility: If bidder with value v_i bids as if value w :

$$U(w; v_i) = v_i \cdot F^{n-1}(w) - B(w) \quad \text{Setup}$$

$$\frac{\partial U}{\partial w} = v_i(n-1)f(w)F^{n-2}(w) - B'(w) = 0 \quad \text{FOC}$$

$$B'(v_i) = v_i(n-1)f(v_i)F^{n-2}(v_i) \quad (w = v_i)$$

$$B(v_i) = \int_{\underline{v}}^{v_i} t(n-1)f(t)F^{n-2}(t) dt = \int_{\underline{v}}^{v_i} t dF^{n-1}(t) \quad B(\underline{v}) = 0$$

$$B(v_i) = v_i F^{n-1}(v_i) - \int_{\underline{v}}^{v_i} F^{n-1}(t) dt \quad \text{By parts}$$

Uniform Distribution ($F(v) = v$):

$$B(v_i) = \frac{n-1}{n} v_i^n$$

Derivation: All-Pay First-Price Uniform Example

Setup: $v_i \sim U[0, 1]$, so $F(v) = v$

$$B(v_i) = v_i F^{n-1}(v_i) - \int_0^{v_i} F^{n-1}(t) dt \quad \text{From above}$$

$$B(v_i) = v_i \cdot v_i^{n-1} - \int_0^{v_i} t^{n-1} dt \quad F(t) = t$$

$$B(v_i) = v_i^n - \left[\frac{t^n}{n} \right]_0^{v_i}$$

$$\boxed{B(v_i) = \frac{n-1}{n} v_i^n}$$

Key Properties:

- $B(v_i) < v_i$ for all $v_i < 1$ (bid shading is severe)
- $B'(v_i) = (n-1)v_i^{n-1} > 0$ (increasing in value)
- $B(1) = \frac{n-1}{n}$ (highest type bids close to value)
- Expected payment: $\mathbb{E}[B(v)] = \frac{n-1}{n(n+1)}$

What is an All-Pay Second Price Auction

Definition

An **all-pay second price auction** (also called **war of attrition**) is a **simultaneous** market mechanism where the highest bidder wins and pays the **second-highest bid**, but **all losing bidders also pay their own bids**.

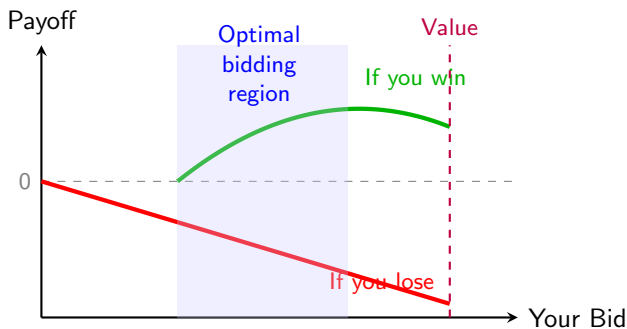
Sealed-Bid All-Pay Second Price Format

- Bidders submit sealed bids
- Highest bid wins the item
- Winner pays the **second-highest bid**
- **All losers pay their own bids**

War of Attrition: Strategic Dynamics

The Unique Challenge

- Winner pays less than their bid (second-price benefit)
- But losers still pay their full bids (all-pay cost)
- Creates complex strategic trade-offs



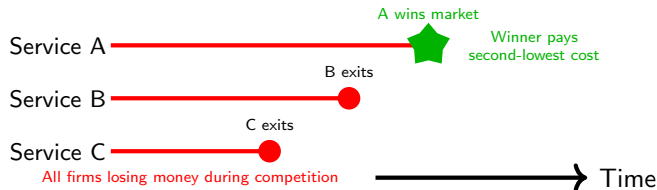
Key Insight: Optimal bid balances: (1) winning probability, (2) payment if winning (2nd price), (3) loss if losing (your bid)

War of Attrition in Business: Market Competition

Real-World Applications

- **Price wars:** Firms sustain losses while competing; survivor gains market
- **Patent races:** All firms pay development costs; winner gets patent protection
- **Standard-setting battles:** All invest in promotion; winner sets industry standard
- **Market entry deterrence:** Incumbents and challengers burn resources

Example: Streaming Service Price War



Derivation: War of Attrition (All-Pay Second-Price)

$$U(w; v_i) = \mathbb{P}(\text{win}) \cdot (v_i - \mathbb{E}[\text{2nd bid} \mid \text{win}]) - \mathbb{P}(\text{lose}) \cdot B(w)$$

With symmetric strategies $B(\cdot)$:

$$U(w; v_i) = F^{n-1}(w) \cdot v_i - \int_{\underline{v}}^w B(t) dF^{n-1}(t) - (1 - F^{n-1}(w))B(w)$$

$$\begin{aligned} \frac{\partial U}{\partial w} &= (n-1)f(w)F^{n-2}(w)v_i - B(w)(n-1)f(w)F^{n-2}(w) \\ &\quad + (n-1)f(w)F^{n-2}(w)B(w) - B'(w)(1 - F^{n-1}(w)) = 0 \end{aligned}$$

Simplify at $w = v_i$:

$$B'(v_i)(1 - F^{n-1}(v_i)) = (n-1)f(v_i)F^{n-2}(v_i)v_i$$

$$B'(v_i) = \frac{(n-1)f(v_i)F^{n-2}(v_i)}{1 - F^{n-1}(v_i)}v_i = xv_i$$

Derivation: War of Attrition ODE Solution

$$B'(v_i) = xv_i; \frac{d}{dv} [-\ln(1 - F^{n-1}(v))] = x$$

$$\rightarrow B'(v_i) = -v_i \frac{d}{dv_i} \ln(1 - F^{n-1}(v_i))$$

$$B(v_i) = -v_i \ln(1 - F^{n-1}(v_i)) + \int_{\underline{v}}^{v_i} \ln(1 - F^{n-1}(t)) dt; B(\underline{v}) = 0$$

$$= -v_i \ln(1 - v_i^{n-1}) + \int_0^{v_i} \ln(1 - t^{n-1}) dt; \textbf{Uniform: } F(v) = v \text{ on } [0, 1]$$

$$= -v_i \ln(1 - v_i) + \int_0^{v_i} \ln(1 - t) dt; \textbf{Two-Player Case: } n = 2$$

$$= -v_i \ln(1 - v_i) - v_i + v_i \ln(1 - v_i) + \int_0^{v_i} \frac{t}{1 - t} dt$$

$$B(v_i) = v_i + (1 - v_i) \ln(1 - v_i)$$

Tullock Contests

Contest Success Function:

$$p_i(x_i, x_{-i}) = \begin{cases} \frac{x_i^r}{\sum_{j=1}^N x_j^r} & \text{if } \sum_j x_j > 0 \\ \frac{1}{N} & \text{otherwise} \end{cases}$$

Parameter r interpretation:

- $r \rightarrow 0$: Random lottery (effort irrelevant)
- $r = 1$: Lottery contest (linear in effort)
- $r \rightarrow \infty$: All-pay auction (deterministic)

Symmetric Equilibrium:

$$x^* = \frac{r(N-1)V}{N^2}$$

Rent Dissipation: Total effort = $\frac{r(N-1)}{N} \times V$
Existence requires: $r \leq 2$ for $N \geq 2$

Optimal Prize Allocation

Moldovanu-Sela (2001): How many prizes?

Setup:

- k contestants with ability $c_i \sim F[m, 1]$ (low c = high ability)
- Cost of effort x_i : $c_i \gamma(x_i)$
- Prizes: $V_1 \geq V_2$, with $V_1 + V_2 = 1$
- Designer maximizes total expected effort

Key Results:

- **Linear/Concave costs:** Single prize optimal
- **Convex costs:** Multiple prizes may be optimal

Intuition:

- First prize: Increases effort of high-ability types
- Second prize: Motivates middle/low-ability types
- With convex costs, second effect can dominate

Derivation: Tullock Contest - Setup

Contest Success Function (CSF):

$$p_i(x_i, x_{-i}) = \begin{cases} \frac{x_i^r}{\sum_{j=1}^n x_j^r} & \text{if } \sum_j x_j > 0 \\ \frac{1}{n} & \text{otherwise} \end{cases}$$

Parameters:

- $x_i \geq 0$: Effort/expenditure by contestant i
- $r > 0$: Decisiveness parameter (technology of conflict)
- V : Prize value (same for all contestants)
- Cost of effort: $c(x_i) = x_i$ (linear cost)

Expected Utility:

$$U_i(x_i, x_{-i}) = p_i(x_i, x_{-i}) \cdot V - x_i = \frac{x_i^r}{\sum_{j=1}^n x_j^r} \cdot V - x_i$$

Goal: Find symmetric Nash equilibrium where all players choose $x^* = x_i = x_j$ for all i, j

Derivation: Tullock Contest - FOC

Expected Utility of Player i :

$$U_i = \frac{x_i^r}{x_i^r + \sum_{j \neq i} x_j^r} \cdot V - x_i$$

In Symmetric Equilibrium: All others choose $x_j = x^*$, so:

$$U_i = \frac{x_i^r}{x_i^r + (n-1)(x^*)^r} \cdot V - x_i$$

First-Order Condition:

$$\frac{\partial U_i}{\partial x_i} = V \cdot \frac{rx_i^{r-1}[x_i^r + (n-1)(x^*)^r] - x_i^r \cdot rx_i^{r-1}}{[x_i^r + (n-1)(x^*)^r]^2} - 1 = 0$$

Simplify:

$$\frac{\partial U_i}{\partial x_i} = V \cdot \frac{rx_i^{r-1} \cdot (n-1)(x^*)^r}{[x_i^r + (n-1)(x^*)^r]^2} - 1 = 0$$

At Symmetric Equilibrium ($x_i = x^*$):

$$V \cdot \frac{r(x^*)^{r-1} \cdot (n-1)(x^*)^r}{[x^* + (n-1)x^*]^2} = 1$$

Derivation: Tullock Contest - Solution

From FOC:

$$V \cdot \frac{r(n-1)(x^*)^{2r-1}}{n^2(x^*)^{2r}} = 1$$

$$\frac{V \cdot r(n-1)}{n^2 \cdot x^*} = 1$$

Solve for Equilibrium Effort:

$$x^* = \frac{r(n-1)V}{n^2}$$

Properties:

- **Comparative Statics:**

- $\frac{\partial x^*}{\partial V} > 0$: Higher prize $\Rightarrow$ more effort
- $\frac{\partial x^*}{\partial r} > 0$: More decisive contest $\Rightarrow$ more effort
- $\frac{\partial x^*}{\partial n} = \frac{rV(n-2)}{n^3}$: Non-monotonic in n

- **Total Effort:** $n \cdot x^* = \frac{r(n-1)V}{n}$

- **Rent Dissipation:** $\frac{n \cdot x^*}{V} = \frac{r(n-1)}{n} \rightarrow r$ as $n \rightarrow \infty$

Existence Condition: Second-order condition requires $r < 2$ for $n > 2$

Derivation: Tullock Contest - Special Cases

Equilibrium Effort: $x^* = \frac{r(n-1)V}{n^2}$

Case 1: All-Pay Auction ($r \rightarrow \infty$)

$$\lim_{r \rightarrow \infty} \frac{r(n-1)V}{n^2} = \infty$$

Contest becomes deterministic: highest effort wins with certainty

Case 2: Pure Lottery ($r \rightarrow 0$)

$$\lim_{r \rightarrow 0} \frac{r(n-1)V}{n^2} = 0$$

Effort doesn't matter: each player has probability $\frac{1}{n}$ regardless

Case 3: Linear Contest ($r = 1$)

$$x^* = \frac{(n-1)V}{n^2}$$

Case 4: Two Players ($n = 2$)

$$x^* = \frac{rV}{4}$$

Rank-Order Tournaments

Lazear-Rosen (1981): Tournament vs. Piece Rates

Output: $y_i = f(e_i) + \epsilon_{common} + \epsilon_i$

Key Insights:

- Risk-neutral agents: All schemes equivalent
- Risk-averse agents:
 - Large common shock $\Rightarrow$ Tournament better
 - Large idiosyncratic shock $\Rightarrow$ Piece rates better

Green-Stokey (1983): Information Value

- Tournaments filter common shocks via relative performance
- Large groups: Ranking $\approx$ precise performance measure
- Useful when common shock distribution unknown

Practical advantages:

- Easier to measure ranks than absolute output
- Robust to non-stationary environments

Elimination Tournaments

Rosen (1986): Sequential Contests

Key Finding: Need disproportionately large top prize

Option Value Logic:

- Each stage: Winner gets option to continue
- As stages progress, fewer opportunities remain
- Final stage: No tomorrow $\Rightarrow$ option expires
- Top prize must compensate for all expired options

Behavioral Effects:

- Pessimists: Underperform (appear weaker, value lower)
- Optimists: Mixed (underestimate rivals vs. overvalue continuation)
- Signaling incentives increase with prize concentration

Application: Executive compensation ladders

Application: Litigation Systems

Baye-Kovenock-de Vries (2005): Fee-Shifting Rules Legal Systems:

- USA: Each pays own ($\alpha = \beta = 1$)
- UK: Loser pays all ($\alpha = 1, \beta = 0$)
- Continental: Mixed ($\alpha = 1, \beta \in (0, 1)$)

Equilibrium Expenditure:

$$e(v) = \frac{\int_0^v t[\alpha - (\alpha - \beta)F(t)]dF(t)}{[\alpha - (\alpha - \beta)F(v)]^2}$$

Results:

- Per-trial costs: UK > Continental > USA
- Settlement incentives: USA > Continental > UK
- Total litigation: USA > Continental > UK

Key: USA system encourages more cases but lower per-case spending

Application: Status Competition

Hopkins-Kornienko (2004, 2006): Growth and Inequality

Status concerns in consumption:

$$U = \log S(c_i, F(c)) + \rho \log c_{i,t+1}$$

where $S(c_i, F) = S_0 + \delta F(c_i) + (1 - \delta)F^-(c_i)$

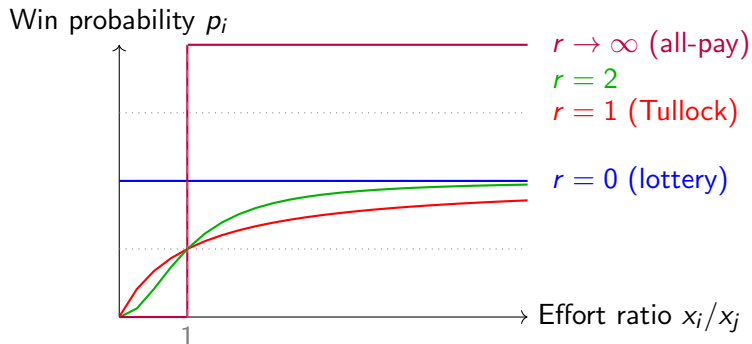
Equilibrium investment:

$$k(z) = \frac{zS_0^\nu + \int_{\underline{z}}^z (S_0 + G(y))^\nu dy}{(S_0 + G(z))^\nu}$$

Key Results:

- "Harsh" competition ($S_0 = 0$): Redistribution $\downarrow$ growth
- "Mild" competition ($S_0 > 0$): Redistribution may $\uparrow$ growth
- Competitiveness effect can dominate scale effect

Contest Success Functions: Visualization



Derivation: Rank-Order Tournament - Setup (Lazear-Rosen 1981)

Model:

- Two risk-averse workers: $i \in \{1, 2\}$
- Output: $q_i = e_i + \varepsilon_i + \theta$
 - e_i : Effort (choice variable)
 - ε_i : Idiosyncratic shock, $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$, independent
 - θ : Common shock, $\theta \sim N(0, \sigma_\theta^2)$
- Prizes: $w_1 > w_2$ (winner and loser prizes)
- Utility: $u(w) - c(e)$ where $u'(w) > 0$, $u''(w) < 0$, $c'(e) > 0$, $c''(e) > 0$
- Cost of effort: $c(e) = \frac{1}{2}e^2$ (quadratic)

Tournament Rule:

Worker i wins if $q_i > q_j \Leftrightarrow e_i + \varepsilon_i > e_j + \varepsilon_j$

Goal: Derive optimal prize spread (w_1, w_2) and equilibrium effort

Derivation: Rank-Order Tournament - Worker's Problem

Symmetric Equilibrium: Both workers choose same effort $e^* = e_1 = e_2$

Worker i 's Winning Probability:

$$p_i = \mathbb{P}(q_i > q_j) = \mathbb{P}(e_i + \varepsilon_i + \theta > e_j + \varepsilon_j + \theta)$$

$$= \mathbb{P}(e_i + \varepsilon_i > e_j + \varepsilon_j) = \mathbb{P}(\varepsilon_i - \varepsilon_j > e_j - e_i)$$

In symmetric equilibrium ($e_i = e_j = e^*$):

$$p_i = \mathbb{P}(\varepsilon_i - \varepsilon_j > 0) = \frac{1}{2}$$

Expected Utility:

$$EU_i(e_i | e_j = e^*) = p_i(e_i, e^*) \cdot u(w_1) + (1 - p_i(e_i, e^*)) \cdot u(w_2) - c(e_i)$$

Note: $\varepsilon_i - \varepsilon_j \sim N(0, 2\sigma_\varepsilon^2)$, so:

$$p_i(e_i, e_j) = \Phi\left(\frac{e_i - e_j}{\sqrt{2\sigma_\varepsilon^2}}\right) = \Phi\left(\frac{e_i - e_j}{\sigma_\varepsilon \sqrt{2}}\right)$$

where $\Phi(\cdot)$ is the standard normal CDF vation: War of Attrition (All

Derivation: Rank-Order Tournament - FOC

Expected Utility:

$$EU_i = \Phi\left(\frac{e_i - e^*}{\sigma_\varepsilon \sqrt{2}}\right) u(w_1) + \left[1 - \Phi\left(\frac{e_i - e^*}{\sigma_\varepsilon \sqrt{2}}\right)\right] u(w_2) - \frac{1}{2}e_i^2$$

First-Order Condition:

$$\frac{\partial EU_i}{\partial e_i} = \phi\left(\frac{e_i - e^*}{\sigma_\varepsilon \sqrt{2}}\right) \cdot \frac{1}{\sigma_\varepsilon \sqrt{2}} \cdot [u(w_1) - u(w_2)] - e_i = 0$$

where $\phi(\cdot)$ is the standard normal PDF

At Symmetric Equilibrium ($e_i = e^* = e_j$):

$$\phi(0) \cdot \frac{1}{\sigma_\varepsilon \sqrt{2}} \cdot [u(w_1) - u(w_2)] = e^*$$

Since $\phi(0) = \frac{1}{\sqrt{2\pi}}$:

$$\frac{1}{\sqrt{2\pi}} \cdot \frac{1}{\sigma_\varepsilon \sqrt{2}} \cdot [u(w_1) - u(w_2)] = e^*$$

Equilibrium Effort:

Derivation: Tournament vs Piece-Rate

Piece-Rate Compensation: $w_i = \beta q_i + \alpha$

$$EU_i^{PR} = \mathbb{E}[u(\beta(e_i + \varepsilon_i + \theta) + \alpha)] - c(e_i)$$

Risk-Neutral Case:

- Tournament: Effort depends on $\Delta u = u(w_1) - u(w_2)$ and σ_ε
- Piece-rate: Effort depends on β
- Both equivalent: Can set Δu to induce same effort as optimal β

Risk-Averse Case with Shocks:

Large Common Shock (σ_θ^2 large):

- **Tournament better:** Common shock cancels in relative performance
- Worker bears less risk in tournament
- $q_i - q_j = e_i - e_j + \varepsilon_i - \varepsilon_j$ (no θ !)

Large Idiosyncratic Shock (σ_ε^2 large):

- **Piece-rate better:** Tournament adds noise to compensation
- High-performing worker may lose due to bad luck
- From FOC: $e^* \propto \frac{1}{\sigma_\varepsilon}$ (effort decreases with noise)

Derivation: Optimal Prize Spread

Principal's Problem: Choose (w_1, w_2) to maximize profits subject to:

- ① Incentive Compatibility (IC): Worker chooses optimal effort
- ② Individual Rationality (IR): Worker accepts contract

Constraints:

$$e^* = \frac{u(w_1) - u(w_2)}{\sigma_\varepsilon \cdot 2\sqrt{\pi}} \quad (\text{IC})$$

$$\frac{1}{2}u(w_1) + \frac{1}{2}u(w_2) - c(e^*) \geq \bar{u} \quad (\text{IR})$$

Principal's Profit:

$$\Pi = \mathbb{E}[q_1 + q_2] - (w_1 + w_2) = 2e^* - w_1 - w_2$$

Lagrangian:

$$\mathcal{L} = 2e^* - w_1 - w_2 + \lambda \left[\frac{1}{2}u(w_1) + \frac{1}{2}u(w_2) - c(e^*) - \bar{u} \right]$$

subject to IC constraint

Key Result: Optimal prize spread balances:

- Higher spread $\Rightarrow$ more effort (good)

Derivation: Risk-Neutral Benchmark

If Workers Are Risk-Neutral: $u(w) = w$

Equilibrium Effort:

$$e^* = \frac{w_1 - w_2}{\sigma_\varepsilon \cdot 2\sqrt{\pi}}$$

Principal's Problem:

$$\begin{aligned} \max_{w_1, w_2} \quad & 2e^* - w_1 - w_2 \\ \text{s.t.} \quad & \frac{1}{2}w_1 + \frac{1}{2}w_2 - \frac{1}{2}(e^*)^2 \geq \bar{u} \end{aligned}$$

From IC: $w_1 - w_2 = e^* \cdot \sigma_\varepsilon \cdot 2\sqrt{\pi}$

FOC with respect to e^* :

$$2 = \lambda e^* + \text{shadow price from IC}$$

Solution: Set prize spread to induce first-best effort e^{FB} where:

$$\text{Marginal product} = \text{Marginal cost} \Rightarrow 1 = c'(e^{FB}) = e^{FB}$$

$$e^* = 1, \quad w_1 - w_2 = \sigma_\varepsilon \cdot 2\sqrt{\pi}$$

Summary: Contests and Tournaments

Tullock Contest:

- Equilibrium effort: $x^* = \frac{r(n-1)V}{n^2}$
- Rent dissipation: $\frac{r(n-1)}{n}$ fraction of prize
- More decisive technology (r higher) $\Rightarrow$ higher effort
- Inverted-U in number of contestants

Rank-Order Tournament:

- Equilibrium effort: $e^* = \frac{u(w_1) - u(w_2)}{\sigma_\varepsilon \cdot 2\sqrt{\pi}}$
- Optimal prize spread balances incentives vs. risk
- Tournament dominates piece-rate when common shocks large
- Piece-rate dominates when idiosyncratic shocks large

Key Insights:

- Contest technology (r) critically affects equilibrium behavior
- Relative performance evaluation filters common shocks
- Optimal mechanism depends on noise structure

Summary: Bidding Functions - All Formats

Uniform Distribution $v \sim U[0, 1]$, n bidders:

Auction Format	Equilibrium Bid $B(v)$	Revenue
English	v	$\mathbb{E}[v_{(2)}] = \frac{n-1}{n+1}$
Second-Price	v	$\mathbb{E}[v_{(2)}] = \frac{n-1}{n+1}$
First-Price	$\frac{n-1}{n} v$	$\mathbb{E}[v_{(2)}] = \frac{n-1}{n+1}$
Dutch	$\frac{n-1}{n} v$	$\mathbb{E}[v_{(2)}] = \frac{n-1}{n+1}$
All-Pay 1st	$\frac{n-1}{n} v^n$	$\mathbb{E}[v_{(2)}] = \frac{n-1}{n+1}$
War of Attrition	$v + (1 - v) \ln(1 - v)$ (for $n = 2$)	$\mathbb{E}[v_{(2)}] = \frac{n-1}{n+1}$

Key Observations:

- Revenue Equivalence holds under IPV, risk neutrality, symmetry
- Bid shading increases with: competition (larger n), all-pay structure
- Truth-telling optimal only in English and Second-Price formats

Benchmark Model Setup

Three Key Assumptions:

① Independent Private Values (IPV):

- Valuations statistically independent
- Learning others' values doesn't change own value

② Risk Neutrality:

- Utility $U_i = v_i - b_i$ if win, 0 if lose
- No risk aversion or risk seeking

③ Symmetry:

- All bidders from same distribution F
- No ex-ante differences between bidders

Notation:

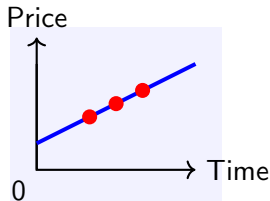
- n bidders, valuations $v_i \sim F(v)$ on $[\underline{v}, \bar{v}]$
- Density $f(v) = F'(v)$, assume $f > 0$ on support
- Order statistics: $v_{(1)} \geq v_{(2)} \geq \dots \geq v_{(n)}$
- Seller's valuation: $v_s = 0$ (for simplicity)

Strategic and Outcome Equivalence

Outcome Equivalence

**Ascending
Auction**

**Second Price
Sealed Bid**

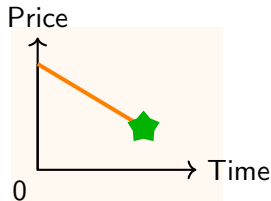


Bidders drop out

Strategic Equivalence

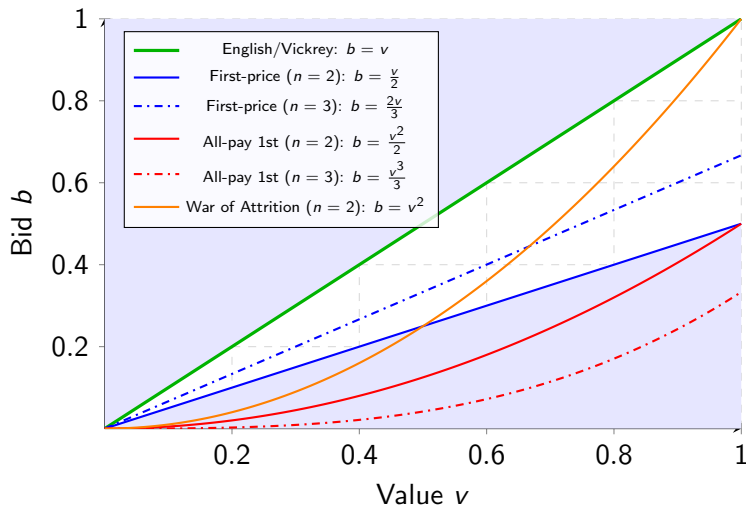
**Descending
Auction**

**First Price
Sealed Bid**



First to accept wins

Visualization: Bid Functions Across Auction Types

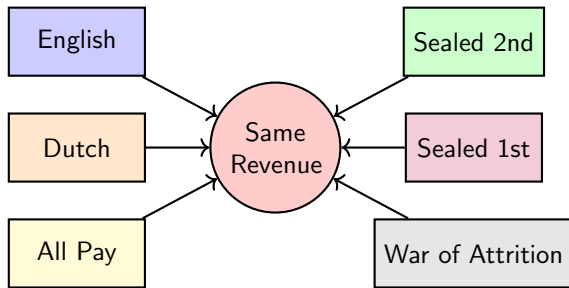


Key insight 1: All-pay auctions have more aggressive bid shading

Key insight 2: Mechanism determines effect of competition

The Big Surprise Revenue Equivalence

Amazing Result Under standard conditions, all four formats generate the same expected revenue!



Conditions Required

- Private values (each bidder knows their own value)
- Risk-neutral bidders
- No collusion
- Symmetric bidders (draw from same distribution)

Implication Choose format based on practical considerations, not revenue

Expected Revenue Calculation

General Formula:

$$ER = \int_{\underline{v}}^{\bar{v}} [\text{payment by winner}] \cdot [\text{probability density}] dv$$

English & Vickrey: Winner pays $v_{(2)}$

$$ER_{SPA} = \mathbb{E}[v_{(2,n)}] = \int_{\underline{v}}^{\bar{v}} v \cdot g_{(2)}(v) dv$$

where $g_{(2)}$ is density of second-order statistic

First-Price & Dutch: Winner pays $B(v_{(1)})$

$$ER_{FPA} = \int_{\underline{v}}^{\bar{v}} B(v) \cdot g_{(1)}(v) dv = \mathbb{E}[v_{(2,n)}]$$

(Can prove using integration by parts)

Uniform Example: Both equal $\frac{n-1}{n+1}$

Revenue Equivalence Theorem (RET)

Theorem (Vickrey 1961, Myerson 1981, Riley-Samuelson 1981)

Consider any two auction mechanisms satisfying:

- 1 *Bidders are risk-neutral with i.i.d. private values from F*
- 2 *In equilibrium, object goes to bidder with highest value*
- 3 *Bidder with lowest value $\underline{v}$ has zero expected surplus*

Then both mechanisms yield the same expected revenue to the seller.

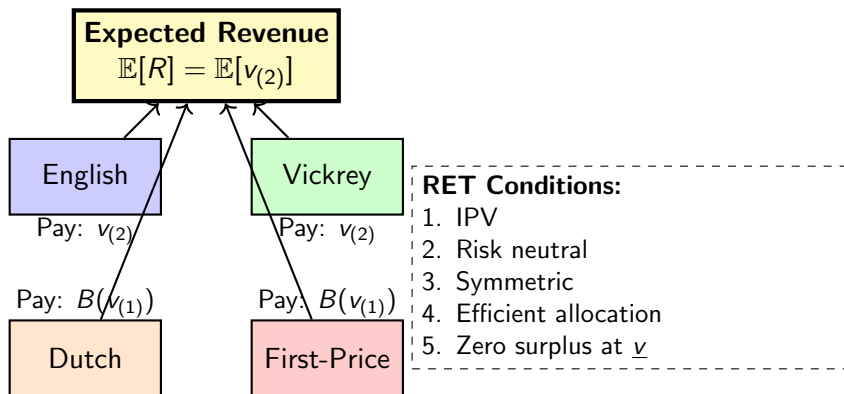
Remarkable Implication:

$$ER_{\text{English}} = ER_{\text{Vickrey}} = ER_{\text{Dutch}} = ER_{\text{First-Price}} = \mathbb{E}[v_{(2)}]$$

Intuition:

- Revenue determined by competition level, not payment rule
- Bidders adjust strategies to auction format
- In equilibrium, same economic forces prevail

Visualization: Revenue Equivalence Intuition



The Auction Trilemma You Can't Have It All

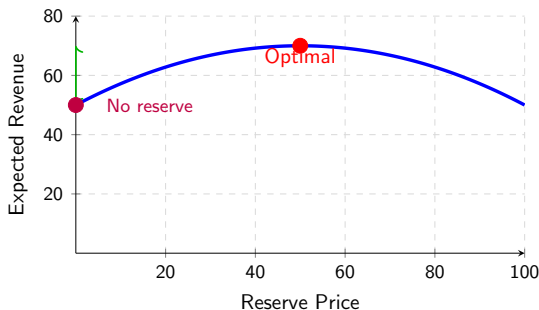
The Three Desirable Properties



Figure: An auction trilemma **Static** no time, all at once. **Strategy proof** no reason to bid anything other than your true value. **Credible** Auctioneer has no strategy to secretly get more revenue. Can't have all three

Reserve Prices Almost Always Use Them

Reserve Price = Minimum acceptable bid



Why They Work

- Forces bidders to compete more aggressively
- Protects against low-ball bids
- New Zealand spectrum disaster no reserve = \$1 winning bid!

Entry Decisions and Participation

Endogenous Entry:

- Entry cost $c > 0$ to participate
- Symmetric equilibrium: enter with probability p^*
- Expected number of bidders: $n^* = Np^*$

Optimal Auction with Entry:

- Reserve price affects entry incentives
- Too high r : Discourages entry
- Too low r : Excessive entry (wasteful)
- Optimal r balances revenue vs. entry

Entry Fees:

- Can screen out low-value bidders
- Revenue from fees + auction proceeds
- Optimal fee extracts entry rent

Collusion and Bid Rigging

Forms of Collusion:

- Bid suppression: Some bidders don't bid
- Bid rotation: Take turns winning
- Cover bidding: Submit intentionally losing bids

Factors Facilitating Collusion:

- Repeated interaction
- Small number of bidders
- Observable bids (English auction)
- Stable market conditions

Anti-Collusion Mechanisms:

- Sealed-bid formats (harder to coordinate)
- Reserve prices (limit collusive gains)
- Random participation rules
- Legal penalties and monitoring

Information Disclosure Policies

What to Reveal?

- Number of bidders
- Identities of participants
- Past winning bids
- Reserve price

Trade-offs:

More Information	Less Information
↑ Reduces uncertainty	↑ Prevents collusion
↑ Encourages participation	↑ Strategic ambiguity
↓ May facilitate collusion	↓ May discourage entry

Optimal Policy:

- Depends on market characteristics
- Common values: More info generally better
- Private values with collusion risk: Less info better

Summary: Core Results

1. Equilibrium Strategies:

- English/Vickrey: Bid true value (dominant strategy)
- First-Price/Dutch: Shade bid below value
- Shading depends on competition and distribution

2. Revenue Equivalence (IPV, risk-neutral, symmetric):

- All standard formats yield same expected revenue
- Revenue = $\mathbb{E}[v_{(2)}]$ (expected second-highest value)
- Breaks with risk aversion, asymmetry, or correlation

3. Optimal Design:

- Reserve price improves revenue
- Virtual valuation determines optimal allocation
- Standard auction + reserve achieves optimum

Summary: Key Mathematical Results

Auction Type	Bid Strategy	Expected Revenue
English	$b = v$	$\mathbb{E}[v_{(2)}]$
Vickrey	$b = v$	$\mathbb{E}[v_{(2)}]$
First-Price	$b = \mathbb{E}[v_{(2)} v_{(1)} = v]$	$\mathbb{E}[v_{(2)}]$
Dutch	$b = \mathbb{E}[v_{(2)} v_{(1)} = v]$	$\mathbb{E}[v_{(2)}]$

Key Formulas:

- Optimal reserve: $r^* = v_s + \frac{1-F(r^*)}{f(r^*)}$
- Virtual valuation: $\psi(v) = v - \frac{1-F(v)}{f(v)}$
- First-price bid (uniform): $B(v) = \frac{n-1}{n} v$
- Risk averse bid: $B(v) = \frac{n-1}{n-(1-\gamma)} v$

Key Insights for Practice

Choosing an Auction Format:

Use English/Ascending when:

Common value elements
Want price discovery
Affiliation present
Transparency important

Use Sealed-Bid when:

Preventing collusion important
Speed matters
Bidders geographically dispersed
Privacy concerns

Design Recommendations:

- Always set optimal reserve price
- Reveal information to reduce winner's curse
- First-price better with risk-averse bidders
- Consider entry fees and participation costs
- Multi-unit: Be aware of demand reduction

Remember: Real auctions rarely satisfy all RET assumptions!

Pure Common Value Model

Setup:

- True value V unknown to all at auction time
- Each bidder i observes private signal S_i
- Signals correlated with true value
- Classic example: Oil drilling rights

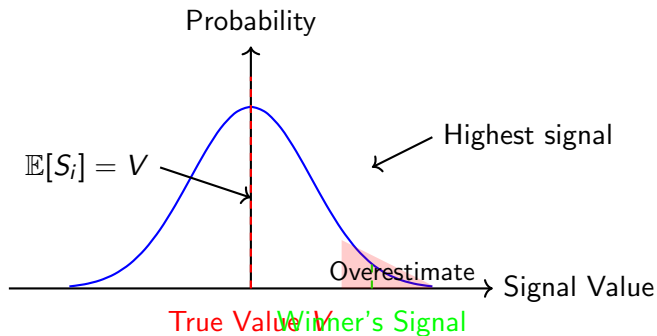
The Winner's Curse:

- Winning means you had the highest signal/estimate
- If signals unbiased: $\mathbb{E}[S_i] = V$
- But: $\mathbb{E}[V|S_i = s, \text{Win}] < s$ typically
- Naive bidding (based on $\mathbb{E}[V|S_i]$) leads to losses!

Rational Response:

- Account for adverse selection of winning
- Bid conditional on being “just winning”
- Shade bid more with more competitors

Visualizing the Winner's Curse



Key Insight: The winner is the bidder with the most optimistic signal

- Even with unbiased signals, winning is “bad news”
- Must adjust bid downward to account for this selection effect

Simple Common Value Example

Uniform Signal Model:

- Common value V
- Signals $t_i \sim U[V - \delta, V + \delta]$ independently
- $\mathbb{E}[V|t_i] = t_i$ (unbiased)
- $\mathbb{E}[V|\text{all signals}] = \frac{t_{(1)} + t_{(n)}}{2}$

Equilibrium Strategies:

Auction	Bid Function	Winner Pays
Second-Price	$b(t_i) = t_i - \delta \frac{n-2}{n}$	Second-highest bid
English	Update as others drop	$\frac{t_{(2)} + t_{(n)}}{2}$
First-Price	$b(t_i) = t_i - \delta$	Own bid

Note: All bid below signal to avoid winner's curse!

Winner's Curse: Theory and Evidence

Rational Response in Common Value Auctions:

Bidder i with signal s_i should bid:

$$b(s_i) = \mathbb{E}[V | S_i = s_i, \max_{j \neq i} S_j = s_i] < \mathbb{E}[V | S_i = s_i]$$

Experimental Evidence (Kagel & Levin):

- Persistent overbidding even with experience
- Bankruptcy rates: 40-60% in lab experiments
- Worse with more bidders (stronger adverse selection)

Real-World Examples:

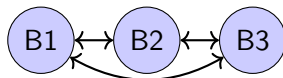
- **Oil lease auctions:** Average returns near zero for winners
- **Corporate takeovers:** Acquiring firm stock typically falls
- **Construction contracts:** Lowest bidder often loses money

Mitigation Strategies

Joint bidding, Information acquisition, Conservative bidding rules

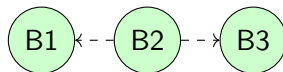
Information Flow in Different Formats

English Auction



Full information flow

Second-Price



Winner learns 2nd bid

First-Price



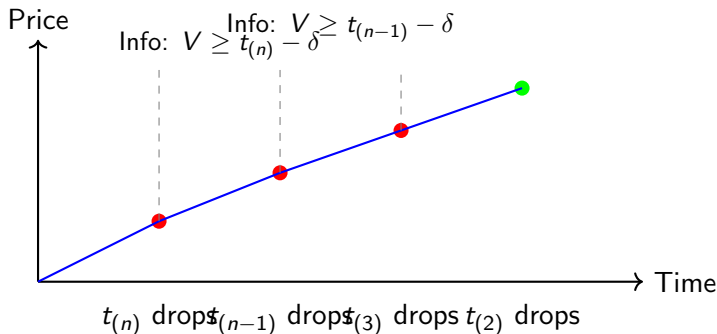
No information flow

Information Revelation Impact:

- More information → Less winner's curse
- Less winner's curse → More aggressive bidding
- More aggressive bidding → Higher revenue

English Auction Information Dynamics

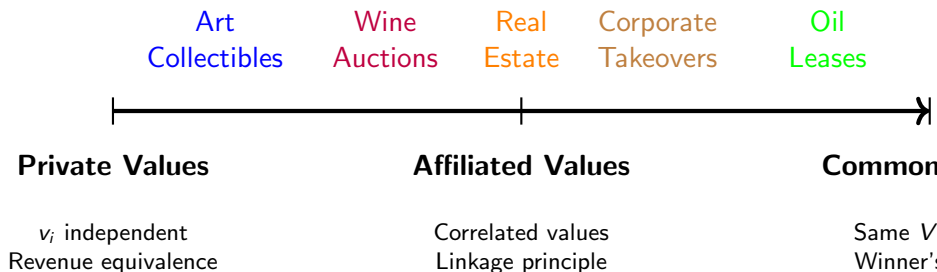
Progressive Information Revelation:



Belief Updates:

- Each dropout reveals that bidder's signal
- Remaining bidders update: $\mathbb{E}[V|\text{history}]$ increases
- Final two bidders have nearly complete information

Value Interdependence Spectrum



Mathematical Representation:

$$v_i = \alpha \cdot (\text{private component}) + (1 - \alpha) \cdot (\text{common component})$$

where $\alpha \in [0, 1]$ determines position on spectrum

Affiliation: Formal Definition

Affiliated Random Variables: Variables $(X_1, \dots, X_n)$ are affiliated if:

$$f(x \vee x') \cdot f(x \wedge x') \geq f(x) \cdot f(x')$$

where $\vee$ = component-wise max, $\wedge$ = component-wise min

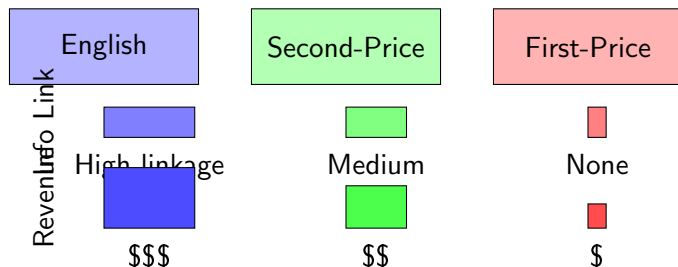
Intuition:

- Higher values of one variable make higher values of others more likely
- Positive correlation, but stronger
- Common in auction settings

Examples:

- Signals about common value
- Expert evaluations of artwork
- Geological surveys for oil

The Linkage Principle Visualized



Milgrom-Weber Revenue Ranking:

$$\mathbb{E}[\text{Revenue}]_{\text{English}} \geq \mathbb{E}[\text{Revenue}]_{\text{Second-Price}} \geq \mathbb{E}[\text{Revenue}]_{\text{First-Price}}$$

Key Insight: Linking winner's payment to others' information reduces winner's information rent

Linkage Principle: Policy Implications

Seller Should Reveal Information:

- Commit to honestly reveal all information
- Reduces winner's curse, more aggressive bidding
- Increases expected revenue
- Even if information is bad!

Examples:

- Art: Provide authentication, condition reports
- Real estate: Home inspections, disclosures
- Oil rights: Share geological surveys
- Companies: Due diligence data rooms

Reserve Price Policy:

- Public reserve better than secret reserve
- Provides information about seller's valuation
- Links payment to seller's information

The Wallet Game (Klemperer 1998)

Model “Almost Common Values”:

$$v_i = \alpha t_i + \beta \sum_{j \neq i} t_j$$

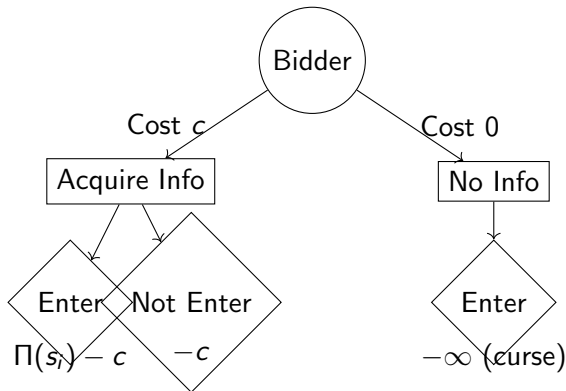
- $\alpha = \beta$: Pure common value (sum of all signals)
- $\beta = 0$: Pure private value
- $\alpha > \beta$: Mostly private with common element

English Auction Equilibrium:

- First dropout: $p_1 = (\alpha + (n-1)\beta)t_{(n)}$
- Information revealed at each dropout
- Final price: $p = \beta \sum_{j=3}^n t_{(j)} + (\alpha + \beta)t_{(2)}$

Key Result: Small advantage (α slightly larger) creates large competitive advantage in English auction with common values

Entry and Information Acquisition



Equilibrium Effects:

- Information acquisition creates entry barrier
- Uninformed bidders face severe winner's curse
- Can lead to thin markets or market breakdown
- English auctions partially mitigate through revelation

Summary: Core Concepts

① Winner's Curse

- Arises in common value settings
- Winning is “bad news” about value
- Rational bidders must shade bids accordingly

② Information Revelation

- English auctions reveal information progressively
- Reduces uncertainty and winner's curse
- Enables more aggressive bidding

③ Linkage Principle

- Link payment to others' information
- Reduces information rents
- English $\geq$ Second-Price $\geq$ First-Price

④ Affiliation

- Stronger than correlation
- Common in real auctions
- Breaks revenue equivalence

Key Takeaways for Practice

For Sellers

- Use English auctions when values are affiliated/common
- Commit to reveal all information (even bad news)
- Set public rather than secret reserve prices
- Provide inspection opportunities and documentation

For Bidders

- Account for winner's curse in common value settings
- Bid more conservatively with more competitors
- Value information acquisition opportunities
- Update beliefs in ascending auctions

For Mechanism Designers

- Consider information structure when choosing format

Mathematical Summary

Key Equilibrium Conditions:

Common Value Second-Price

$$b(s_i) = \mathbb{E}[V | S_i = s_i, S_{(n-1)} = s_i]$$

Revenue Ranking with Affiliation

$$R_{English} = \mathbb{E}[v_{(2)}(S_{(2)}, S_{(2)}, S_{(3)}, \dots, S_{(n)})]$$

$$R_{Second} = \mathbb{E}[v_{(2)}(S_{(2)}, S_{(2)}, S_{(2)}, \dots, S_{(2)})]$$

$$R_{First} = \mathbb{E}[v_{(2)}(S_{(2)}, S_{(1)}, S_{(1)}, \dots, S_{(1)})]$$

Linkage Principle

Higher correlation between payment and others' signals $\Rightarrow$ Higher expected revenue

Multi-Unit Auctions

Setting: k identical units, n bidders, each wants one unit

Uniform Price Auction:

- All winners pay same price (e.g., $(k + 1)$ -th highest bid)
- Strategic: Demand reduction incentive
- Used in Treasury bill auctions

Discriminatory (Pay-as-bid) Auction:

- Each winner pays own bid
- Similar to first-price auction analysis
- Also used in some Treasury auctions

Vickrey-Clarke-Groves (VCG):

- Generalization of second-price auction
- Each pays externality imposed on others
- Efficient but complex, rarely used

Multi-Unit Auction Formats

Setting: K identical units, n bidders, each wants multiple units

1. Discriminatory (Pay-as-Bid) Auction:

- Submit bid schedule $(q_i, b_i(q))$ for quantities
- Winners pay own bids for units won
- Strategic bid shading on all units
- Used in some Treasury bill auctions

2. Uniform Price Auction:

- All winners pay same market-clearing price
- Price = highest rejected bid or lowest accepted
- Demand reduction problem (strategic underbidding)
- Used in electricity markets, IPOs

3. Vickrey (Generalized Second-Price):

- Each pays opportunity cost imposed on others
- Truthful bidding is dominant strategy
- Complex to understand and implement
- Rarely used in practice

Demand Reduction Problem

Issue in Uniform Price Auctions:

Bidding on marginal unit affects price for ALL units won

Example: 2 units available, 2 bidders

- Bidder 1 values: $v_{11} = 10, v_{12} = 8$
- Bidder 2 values: $v_{21} = 9, v_{22} = 7$
- Efficient: Each gets one unit

But in equilibrium:

- Bidder 2 may bid: $b_{21} = 9, b_{22} < 7$
- Reduces demand to lower price on first unit
- Bidder 1 might win both units (inefficient!)

Consequences:

- Inefficient allocation
- Lower revenue than discriminatory auction possible
- Facilitates collusion

Vickrey Auction for Multiple Units

Payment Rule: Bidder i pays the externality imposed on others:

$$p_i = \sum_{j \neq i} v_j(\text{allocation without } i) - \sum_{j \neq i} v_j(\text{allocation with } i)$$

Example: 2 units, 3 bidders with unit demands

- Values: $v_1 = 10, v_2 = 8, v_3 = 6$
- Allocation: Bidders 1 and 2 win
- Bidder 1 pays: 6 (displaces bidder 3)
- Bidder 2 pays: 6 (displaces bidder 3)
- Bidder 3 pays: 0

Properties:

- Dominant strategy: Report true values
- Efficient allocation
- But: Complex, low revenue possible, not collusion-proof

Summary: Core Results

1. Equilibrium Strategies:

- English/Vickrey: Bid true value (dominant strategy)
- First-Price/Dutch: Shade bid below value
- Shading depends on competition and distribution

2. Revenue Equivalence (IPV, risk-neutral, symmetric):

- All standard formats yield same expected revenue
- Revenue = $\mathbb{E}[v_{(2)}]$ (expected second-highest value)
- Breaks with risk aversion, asymmetry, or correlation

3. Optimal Design:

- Reserve price improves revenue
- Virtual valuation determines optimal allocation
- Standard auction + reserve achieves optimum

Key Insights for Practice

Choosing an Auction Format:

Use English/Ascending when:

Common value elements
Want price discovery
Affiliation present
Transparency important

Use Sealed-Bid when:

Preventing collusion important
Speed matters
Bidders geographically dispersed
Privacy concerns

Design Recommendations:

- Always set optimal reserve price
- Reveal information to reduce winner's curse
- First-price better with risk-averse bidders
- Consider entry fees and participation costs
- Multi-unit: Be aware of demand reduction

Remember: Real auctions rarely satisfy all RET assumptions!

Multi-Unit Auctions: Beyond Single Items

Types of Multi-Unit Auctions:

- **Uniform Price:** All winners pay the same market-clearing price
- **Discriminatory (Pay-as-Bid):** Each winner pays their own bid
- **Vickrey-Clarke-Groves (VCG):** Generalized second-price mechanism

Key Design Challenges:

- 1 **Demand Reduction:** Strategic underbidding on marginal units
- 2 **Market Power:** Large bidders can manipulate clearing price
- 3 **Efficiency vs Revenue:** Trade-off becomes more complex

Treasury Auction Example

US Treasury uses discriminatory format for bills, uniform price for notes/bonds. Annual volume: \$12+ trillion (2023)

Combinatorial Auctions: Package Bidding

The Combinatorial Allocation Problem (CAP):

$$\max_{x_{iS}} \sum_{i \in N} \sum_{S \subseteq M} v_{iS} x_{iS} \quad (1)$$

$$\text{s.t.} \quad \sum_{S: j \in S} \sum_{i \in N} x_{iS} \leq 1, \quad \forall j \in M \quad (2)$$

$$\sum_{S \subseteq M} x_{iS} \leq 1, \quad \forall i \in N \quad (3)$$

$$x_{iS} \in \{0, 1\} \quad (4)$$

Complexity and Bidding Languages:

- **XOR Bids:** “A OR B but not both” - Mutually exclusive packages
- **OR Bids:** “A AND/OR B” - Any combination acceptable
- **OR-of-XORs:** Nested structure for complex preferences

Real-World Applications:

- Spectrum auctions (FCC, Ofcom): \$100+ billion raised
- Procurement (Chile school meals): 20% cost reduction

Assumptions for Revenue Equivalence

- ➊ **Risk Neutrality** Bidders maximize expected surplus (not risk-averse)
- ➋ **Independent Private Values (IPV)** Each bidder knows their own valuation v_i , but valuations are drawn independently from distributions
- ➌ **Symmetry** All bidders draw valuations from the same distribution $F(v)$
- ➍ **Allocation Efficiency** The bidder with the highest valuation wins.
- ➎ **Losers Pay Zero** Bidders who don't win pay nothing
- ➏ **Boundary Condition** Expected payment is zero when valuation is zero $E[\text{payment} \mid v_i = 0] = 0$

Key insight Any auction mechanism satisfying these assumptions yields the same expected revenue.

Relaxing Risk Neutrality

The Assumption: Bidders maximize expected monetary value (risk-neutral)

Reality: Most bidders are risk-averse

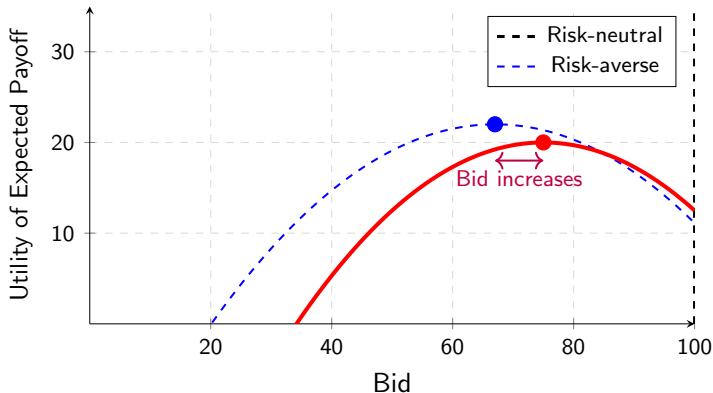
- Prefer certain \$50 over 50% chance of \$100
- Utility function $U(x)$ is concave: $U''(x) < 0$
- Common measures: constant relative risk aversion (CRRA), constant absolute risk aversion (CARA)

Impact on Standard Formats:

- **English & Second-Price:** No impact!
 - Bidding true value still dominant strategy
 - Risk preferences don't affect optimal bid
 - Revenue unchanged
- **First-Price & Dutch:** Higher bids!
 - Risk-averse bidders bid *closer to true value*
 - Trade-off: Lower win probability vs. avoiding risk of not winning
 - **Higher revenue for seller**
 - Efficiency still preserved (highest value still wins)

Risk Aversion in First-Price Auctions: Why Bid Higher?

Intuition: Risk-averse bidders dislike uncertainty



Key Insight: Risk aversion shifts optimal bid higher because:

- Risk-averse bidder values *certainty* of winning
- Willing to sacrifice some profit to increase win probability
- Acts as "competition substitute" - higher bids even with same

Risk Aversion: Revenue Rankings Change

Revenue Equivalence Breaks Down!

With Risk-Neutral Bidders:

First-Price Revenue = Second-Price Revenue = English Revenue

With Risk-Averse Bidders:

First-Price Revenue > Second-Price Revenue = English Revenue

Why This Happens:

- In second-price/English: Bidding true value still optimal
- In first-price: Risk aversion increases bids above risk-neutral level
- **Practical implication:** Sellers benefit from first-price when bidders are risk-averse

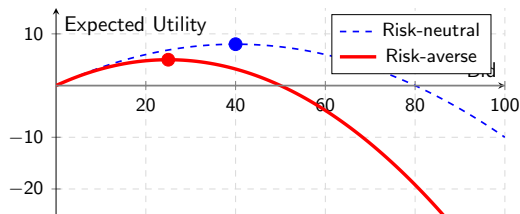
Efficiency:

- **Preserved in all formats!**
- Highest-value bidder still wins (just pays more in first-price)

All-Pay Auctions with Risk Aversion

The Dangerous Combination: Risk aversion + pay even if you lose Strategic Effects:

- Risk-averse bidders *extremely* conservative in all-pay auctions
- Two sources of risk:
 - ① Risk of losing and paying your bid (standard all-pay risk)
 - ② Risk aversion amplifies the pain of losing
- **Result:** Much lower bids than risk-neutral all-pay auction
- **Revenue:** Dramatically lower for seller



Risk aversion + all-pay = very low bidding and participation

Relaxing Symmetry

The Assumption: All bidders draw values from the same distribution $F(v)$

Reality: Different bidder types (strong vs. weak, informed vs. uninformed)

- Strong bidders: Values from $F_S(v)$ with higher support
- Weak bidders: Values from $F_W(v)$ with lower support

Impact on Standard Formats:

- **English & Second-Price:** No impact!
 - Still dominant strategy to bid true value
 - Efficient allocation preserved
 - Revenue = second-highest value
- **First-Price & Dutch:** Major changes!
 - Different equilibrium bid functions $B_i(v)$ per type
 - **Efficiency may be lost:** Weak bidder with higher value may lose to strong bidder with lower value
 - Revenue effects ambiguous (could increase or decrease)

Asymmetric Bidders: Setup

Relaxing Symmetry Assumption:

- Different types of bidders
- Type i values drawn from $F_i(v)$
- Each knows own type and others' types
- But not actual realizations

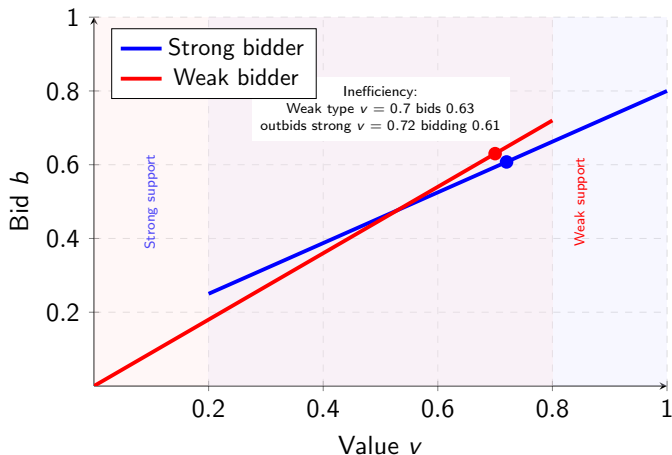
English & Vickrey: No change!

- Still dominant strategy to bid value
- Efficient allocation (highest value wins)
- Revenue = second-highest value

First-Price & Dutch: Major changes!

- Different equilibrium bid functions $B_i(v)$
- May be inefficient: $v_a > v_b$ but $B_i(v_a) < B_j(v_b)$ possible
- No general revenue ranking vs. second-price
- Complex equilibrium computation

Visualization: Asymmetric Equilibrium



Strong: $v \sim U[0.2, 1]$, Weak: $v \sim U[0, 0.8]$

Relaxing Independent Private Values (IPV)

The Assumption: Each bidder's value v_i is independent of others' values

Reality: Common Value or Affiliated Value situations

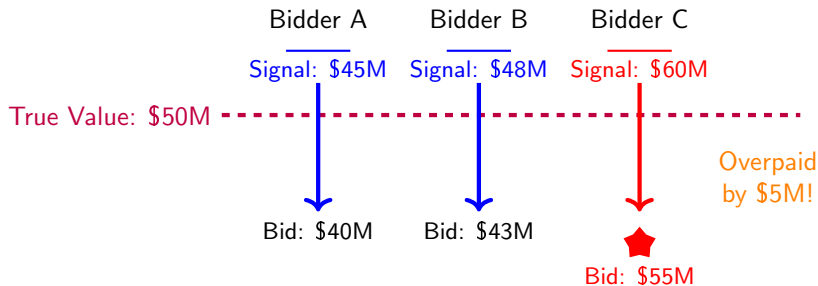
- **Common Values:** True value is the same for all bidders, but each has different information/signal
 - Examples: Oil drilling rights, company acquisitions
- **Affiliated Values:** Values are correlated (if one bidder has high value, likely others do too)
 - Examples: Art, real estate with investment value

Key Problem: The Winner's Curse

- In common value auctions, winning often means you overestimated the value
- Rational bidders bid below their signal to avoid overpaying
- More severe in sealed-bid formats vs. ascending auctions

Winner's Curse: The Danger of Common Values

Scenario: Oil drilling rights with unknown reserves



Lesson: C had the most optimistic (and wrong) signal and won, but overpaid

Impact on Auction Formats: Common Values

English (Ascending) Auction:

- **Advantage:** Information revelation as bidders drop out
- Bidders update beliefs based on others' dropout prices
- Reduces winner's curse severity
- More efficient information aggregation

Sealed-Bid Formats:

- **Disadvantage:** No information revelation
- Winner's curse is more severe
- Bidders shade bids more aggressively
- Lower revenue for seller
- **First-price better than second-price** (paradoxically)

Key Result: Revenue equivalence breaks down!

- English > First-Price > Second-Price (typically)

All-Pay Auctions with Common Values

The Brutal Combination: Winner's curse + everyone pays
Strategic Effects:

- Bidders must be even more conservative
- Risk of overpaying *and* losing money even if you lose
- Very low equilibrium bids
- High variance in outcomes

Real-World Examples:

- **R&D races with uncertain market value**
 - All firms invest (pay), winner gets patent
 - Market value of patent uncertain (common value)
 - Winner often finds market smaller than expected
- **Lobbying for uncertain benefits**
 - All parties spend on lobbying (all pay)
 - Actual benefit of winning uncertain

Moral: All-pay + common values = extremely conservative bidding

Relaxing: More than one relaxation

The Assumption: Only the winner pays; losers pay nothing

Effects When Combined with Other Relaxations:

All-Pay + Risk Aversion:

- Extremely conservative bidding (compounding effect)
- Much lower revenue than standard auctions
- High-value bidders may not participate

All-Pay + Asymmetric Bidders:

- Weak bidders bid even more conservatively
- Strong bidders can bid slightly more aggressively
- Winner selection may be inefficient
- No general formula for equilibrium

Relaxing: Entry Costs and Participation

The Assumption: Free entry, all interested bidders can participate

Reality: Entry costs, qualification requirements, participation barriers

- Qualification fees or deposits
- Information acquisition costs
- Time and effort to prepare bids

Impact on All Formats:

- **Fewer bidders participate**
 - Lower competition → lower revenue
 - Only high-value bidders enter
- **Efficiency may improve or worsen**
 - Better if entry costs filter out low-value bidders
 - Worse if entry costs prevent high-value but resource-constrained bidders
- **Seller trade-off:**
 - High entry fees → fewer bidders but more serious competition
 - Low entry fees → more bidders but potentially less serious

All-Pay Auctions with Entry Costs

The Compounding Effect: Pay to enter + pay your bid regardless of outcome

Strategic Implications:

- Extremely high barrier to participation
- Only bidders with very high expected values enter
- Very aggressive bidding among those who do enter (sunk cost of entry)
- Potential for very few participants

Example: Patent Races with High R&D Setup Costs

- Firms must invest in labs, hire researchers (entry cost)
- Then spend on actual R&D (all-pay aspect)
- Winner gets patent, losers get nothing
- Result: Only 2-3 major firms compete in most drug categories

Seller's Dilemma:

- Want enough bidders for competition
- But each bidder needs high expected surplus to cover entry + bidding costs

Summary: Relaxing Assumptions & All-Pay Formats

Relaxation	Standard Formats	All-Pay Formats
Risk Aversion	First-price: higher bids, same efficiency	Much more conservative bidding
Asymmetry	First-price: may lose efficiency	Weak bidders nearly don't bid
Common Values	English best; 1st > 2nd price	Extremely low bids
Entry Costs	Fewer bidders, trade-off for seller	Very few bidders, high barrier

Key Insight: All-pay auctions are much more sensitive to assumption violations

- Each violation compounds the "pay even if you lose" effect
- Results in very conservative bidding and low participation
- Use all-pay formats only when you truly need them (e.g., modeling contests, lobbying)

Bloomberg Auction Mechanism – Key Features

Mechanism description (to be verified)

- A buyer (or seller) initiates an auction and selects a number of participating counterparties (“offerees”).
- Each offeree submits a bid (or offer) in the form of a buy-sell spread.
- The best (lowest cost) bid wins — i.e. the winner pays/submits their submitted price (first-price style).
- After the auction ends, the winner is informed of the second-best (runner-up) bid.

Applications

- Used in fixed-income or institutional trading environments where transparency and participation are controlled.
- The runner-up disclosure helps the winner gauge their pricing margin and negotiation leverage.

Note The precise implementation details (eligibility, bid format, timing, runner-up disclosure) have ****not**** been fully verified via publicly accessible documentation.

Online Marketplaces eBay vs. Amazon

eBay Model	Amazon Model
English auction format	Fixed price (Buy Now)
Time-limited bidding	Immediate purchase
Price discovery	Price transparency
Seller sets reserve	Seller sets price
Competition visible	Competition hidden
Good for unique items	Good for commodities

Trend Even eBay now emphasizes "Buy It Now" for speed

Lesson Choose mechanism based on

- Product uniqueness
- Buyer urgency
- Transaction costs

Procurement Best Practices

Keys to Success

- **Qualify suppliers first** Don't sacrifice quality for price
- **Clear specifications** Avoid apples-to-oranges comparisons
- **Multiple rounds** Give suppliers chance to compete
- **Reserve your right** Don't have to accept lowest bid

Common Mistakes

- Running auctions with only 2-3 suppliers
- Poorly defined requirements lead to disputes
- Focusing only on price, ignoring total cost
- Burning supplier relationships with aggressive tactics

When to Avoid

- Strategic partnerships or highly customized products
- When you need innovation, not just low price

Building a Marketplace Two-Sided Markets

Your Platform Connects Buyers and Sellers

Examples eBay, Airbnb, Uber, AWS Spot Instances

Key Challenges

- **Chicken-and-egg** Need both sides to participate
- **Pricing** Who pays? How much?
- **Quality and Trust** How to maintain standards?

Auction Design Helps

- **Price discovery** Market finds the right price
- **Matching** Right buyers find right sellers
- **Efficiency** Resources go to highest value use

Critical Decision What's your objective?

- Transaction volume? Revenue? Efficiency? Quality?

Marketplace Design Strategic Choices

Choice	Option A	Option B
Pricing Model	Commission on transactions	Subscription/listing fees
Competition	Winner-take-all	Multiple winners
Transparency	Show all bids/asks	Hide competitor info
Speed	Real-time matching	Batch at intervals
Commitment	Binding immediately	Can cancel/modify

eBay's Choices

- Second-price auctions for trust
- Reputation system for quality
- Buy-it-now for immediacy

Your design choices shape participant behavior

Preventing Collusion

Collusion costs sellers billions annually

Warning Signs	Countermeasures
Identical bids	Use sealed-bid formats
Rotating winners	Set meaningful reserves
Intentionally low bids	Limit information disclosure
Bid withdrawal patterns	Random participation rules
Few active bidders	Encourage new entrants

Best Practice

- Monitor bidding patterns over time
- Use sealed formats when collusion risk is high
- Keep reserve prices confidential
- Prosecute violations aggressively

When Should You Use Auction Design?

Good Fit When

- Selling/buying scarce resources
- Value varies significantly across bidders
- You need price discovery
- You want competition to drive value
- You need transparent, defensible process

Red Flags

- Few potential bidders (easier to collude)
- High complexity (hard to participate)
- Unclear valuation (creates uncertainty)
- High transaction costs

Decision Framework 5 Key Questions

Before designing an auction, ask

1 What are you selling?

- Unique item → English auction
- Commodity → Sealed-bid

2 Who are your bidders?

- Few sophisticated → Watch for collusion
- Many casual → Keep it simple

3 What's your goal?

- Maximum revenue → Set optimal reserve
- Efficiency → English auction

4 How much time do you have?

- Days/weeks → English auction
- Minutes/hours → Sealed-bid

5 What's your risk tolerance?

- Risk-averse → Set higher reserve
- Risk-neutral → Optimize expected value

Implementation Checklist

Before Launching

① Define Success Metrics

- Revenue? Efficiency? Fairness? Speed?

② Know Your Participants

- Sophistication level and collusion potential

③ Set Clear Rules

- Reserve prices, activity requirements, payment terms

④ Plan for Edge Cases

- What if no one bids? Everyone bids same? Manipulation?

Red Flags When Market Design Is Failing

Warning Signs

- **Low participation** Only few bidders showing up
- **Unusual patterns** Everyone bids the same, or very high/low
- **Complaints** Participants say rules are unfair/unclear
- **Gaming** Evidence of manipulation or collusion
- **Instability** Prices swing wildly or predictably
- **Inefficiency** Wrong winners or resources misallocated

What to Do

- Don't ignore the signals
- Gather data and analyze patterns
- Consult with participants
- Be willing to redesign
- Test changes carefully

California ignored warning signs for months before crisis hit

Quick Reference Best Practices

DO

- Set a reserve price (almost always)
- Monitor for collusion patterns
- Keep rules simple and transparent
- Use sealed bids when collusion risk is high

DON'T

- Assume revenue equivalence always holds
- Reveal sensitive competitive information
- Use complex formats without expert help
- Ignore entry barriers and participation costs

The Big Ideas

- ① **Design matters more than theory**
 - Real-world context beats textbook solutions
- ② **Auctions reveal private information**
 - Better than posted prices when values uncertain
- ③ **Format choice depends on context**
 - Under ideal conditions, all formats yield same revenue
 - Choose based on practical considerations
- ④ **Real markets violate ideal conditions**
 - Risk aversion, collusion, entry costs matter
 - Anti-manipulation safeguards essential
- ⑤ **Simplicity beats sophistication**
 - Participants must understand your mechanism
 - Make honesty the dominant strategy

Action Items

Immediate (This Week)

- Review your current procurement processes
- Identify opportunities for competitive bidding

Short-term (This Month)

- Pilot a reverse auction for non-critical spend
- Benchmark against industry practices
- Train team on auction fundamentals

Long-term (This Quarter)

- Assess platform/marketplace opportunities
- Develop auction strategy for key categories
- Build monitoring capabilities

When to Bring in Experts

- High-stakes transactions (\$10M+)
- Complex multi-party negotiations
- New marketplace launches
- Signs of market manipulation

Want to learn more?

• Books for Practitioners

- "Auction Theory" by Vijay Krishna (intermediate)
- "Putting Auction Theory to Work" by Paul Milgrom (advanced)

• Industry Examples

- FCC Spectrum Auctions (www.fcc.gov/auctions)
- Treasury Securities Auctions (www.treasurydirect.gov)
- Google Ads Help Center

• Expert Consultants

- For high-stakes auctions ($\geq \$1M$), hire economists
- Firms NERA, Compass Lexecon, Analysis Group

Questions?